

# Holographic RIS-Aided Reference Energy Detection for Low-Complexity Noncoherent Communications

1<sup>st</sup> Given Name Surname *dept. name of organization (of Aff.)*

*name of organization (of Aff.)*

City, Country

email address or ORCID 2<sup>nd</sup> Given Name Surname *dept. name of organization (of Aff.)*

*name of organization (of Aff.)*

City, Country

email address or ORCID 3<sup>rd</sup> Given Name Surname *dept. name of organization (of Aff.)*

*name of organization (of Aff.)*

City, Country

email address or ORCID

**Abstract**—This paper presents a comprehensive framework for signal recovery in holographic reconfigurable intelligent surface (RIS) receivers, where only amplitude information is available at the receiver. The proposed system employs a known reference wave to enable phase retrieval from intensity-only measurements. We formulate the complete signal model from transmission to reception, and propose a Gerchberg-Saxton algorithm to estimate the wave vector and complex channel gain. Building upon this, we develop a Gauss-Newton algorithm for efficient signal recovery to conduct real-time estimation of the symbols utilizing the property of rapid convergence. Simulation results validate the effectiveness of the proposed approach, showing accurate signal recovery under various signal-to-noise ratio conditions. The complete modeling framework and algorithm implementation details are provided to facilitate reproducibility and further research in RIS-based communication systems.

**Index Terms**—Reconfigurable Intelligent Surface, holographic receiver, phase retrieval, signal recovery, amplitude-only measurements.

## I. INTRODUCTION

Reconfigurable intelligent surfaces (RISs) have emerged as a promising technology for next-generation wireless communication systems, owing to their ability to reshape the wireless propagation environment in a programmable and energy-efficient manner [1], [2]. By adaptively controlling the electromagnetic response of a large number of low-cost reflecting elements, RISs can provide additional degrees of freedom for coverage enhancement, interference management, and spatial signal manipulation. More recently, the functionality of RISs has been extended beyond passive beamforming toward active reception and signal processing, giving rise to the concept of holographic RIS receivers [3]. By exploiting densely deployed metasurface elements, holographic RIS receivers are expected to capture rich spatial information of impinging electromagnetic waves while maintaining much lower hardware complexity than conventional antenna arrays equipped with fully digital radio-frequency chains.

Despite these advantages, a fundamental bottleneck of holographic RIS receivers lies in the acquisition of complex-valued signal information. Conventional coherent receivers rely on down-conversion, local oscillator synchronization, and in-phase/quadrature sampling to obtain both amplitude and phase information. In contrast, practical RIS-based or metasurface-based receivers may only have access to intensity or power measurements due to hardware, power, and integration constraints [4]. As a result, the received signal phase is not directly observable, and the recovery of complex-valued symbols from intensity-only measurements naturally leads to a phase retrieval problem, which has been extensively studied in optics and signal processing [5], [6].

A promising way to alleviate this ambiguity is to introduce a known reference wave before intensity detection. Inspired by optical holography, where a reference beam interferes with an object wave to encode both amplitude and phase information into an intensity pattern [7], a reference-assisted RIS receiver superimposes a known radio-frequency reference signal with the unknown received signal at the metasurface or receiver front end. The resulting square-law measurement contains not only the power of the unknown signal but also a reference-induced interference term. This interference term serves as a phase anchor and makes it possible to infer the hidden complex-valued signal from real-valued intensity observations.

[8] investigated the application of phase retrieval techniques for channel estimation in orthogonal frequency-division multiplexing (OFDM) systems, demonstrating that accurate channel estimation is possible even when only magnitude information is available, which shows the phase retrieval problems often appear in the communication area. [9] proposed an affine phase retrieval framework for wireless communication systems, where the received signal is modeled as an affine transformation of the transmitted signal, and developed algorithms for signal recovery under this model, where the problems formulation is very similar to the reference-assisted phase retrieval model in this paper.

However, existing studies on RIS-assisted reception and

phase retrieval still leave several key questions insufficiently addressed. Most RIS receiver designs focus on enhancing the received signal power, improving beamforming gain, or reducing hardware complexity, while the intrinsic identifiability of complex symbols under intensity-only measurements has not been fully characterized. In particular, when the receiver observes only the squared magnitude of the superimposed field, different transmitted symbols may produce identical or nearly identical intensity patterns. In such cases, no detector, including maximum-likelihood detection or iterative phase retrieval algorithms, can reliably distinguish the transmitted symbols. Therefore, the central issue is not only how to design an efficient recovery algorithm, but also whether the transmitted symbols are uniquely encoded in the intensity domain in the first place.

This issue becomes more critical in holographic RIS receivers because the measured intensity pattern is jointly determined by multiple coupled components, including the reference wave, the pulse-shaping or sampling operation, and the RIS array manifold. If the effective matrix maps two different symbol vectors to the same field response, the corresponding intensity measurements remain indistinguishable regardless of how the detector is implemented. Likewise, even when the effective field responses are different, an improperly designed reference wave may fail to project their phase difference into the observable intensity domain.

Consequently, the lack of a systematic identifiability analysis limits the theoretical understanding and practical design of reference-assisted holographic RIS receivers. Existing phase retrieval methods often assume generic random measurements or focus on continuous signal recovery, whereas communication symbols belong to finite constellations with specific algebraic structures. This discrete nature creates both opportunities and challenges: on one hand, exact recovery may be possible under weaker conditions than those required for general continuous phase retrieval; on the other hand, constellation-dependent ambiguities may still arise if the reference wave and the RIS measurement structure are not properly designed. Therefore, it is essential to investigate how the reference wave, the shaping matrix, and the RIS array manifold jointly determine the uniqueness and stability of symbol recovery.

In this work, we address this problem by developing a reference-assisted phase retrieval framework for holographic RIS receivers. Instead of treating the reference signal merely as an auxiliary hardware component, we characterize its role as a phase-anchoring mechanism that converts hidden phase information into measurable intensity variations. We analyze the discrete identifiability condition of transmitted symbols under the reference-assisted intensity observation model and show that successful recovery requires two complementary properties: the effective RIS measurement matrix must preserve the differences between constellation points, and the reference wave must separate these differences in the intensity domain. Based on this insight, we further introduce an energy-domain minimum distance metric to quantify the distinguishability of different transmitted symbols after square-law detection. This metric provides a direct link between reference-wave design, RIS measurement diversity, and symbol detection

performance.

The main contributions of this paper are summarized as follows:

- We establish a reference-assisted holographic RIS receiver model, in which complex-valued communication symbols are recovered from intensity-only measurements through the interference between the unknown signal and a known reference wave.
- We derive discrete identifiability conditions for reference-assisted phase retrieval over finite constellations and reveal the joint impact of the reference wave, the pulse-shaping matrix, and the RIS array manifold on symbol uniqueness.
- We introduce an energy-domain minimum distance metric to characterize the stability of symbol recovery and to guide reference-wave and RIS measurement design.
- We develop and compare maximum-likelihood and low-complexity phase retrieval based detectors, demonstrating that proper reference-wave diversity significantly improves symbol distinguishability and detection performance.

## II. SYSTEM MODEL

We consider a holographic RIS receiver system in which  $L$  single-antenna users transmit complex baseband symbols to an RIS composed of  $K$  reflecting elements. The RIS senses only the intensity of the interference pattern formed by the received signal and a known reference wave at each of its elements. The known reference signal is generated by a local oscillator [10]. The goal is to estimate the complex channel parameters and degree of arrival (DOA) angles, and subsequently recover the transmitted symbols utilizing these channel information.

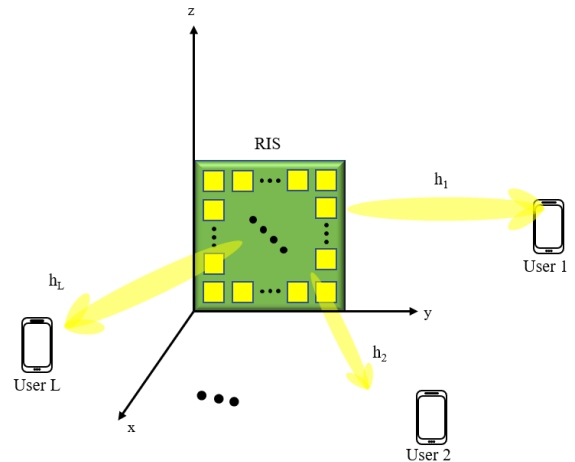


Fig. 1: System model of holographic RIS receiver with amplitude-only measurements.

### A. Transmitter Model

Consider an uplink transmission system with  $L$  single-antenna users. The information-bearing symbols transmitted by the  $l$ -th user within one signaling block are collected as

$\mathbf{s}_l = [s_{l,0}, s_{l,1}, \dots, s_{l,K-1}]^T \in \mathcal{S}^K$ , where  $K$  denotes the number of symbols in the considered block,  $s_{l,k}$  represents the  $k$ -th transmitted symbol of user  $l$ , and  $\mathcal{S}$  denotes the modulation constellation. And the constellation symbols are assumed to be normalized as  $\mathbb{E}\{|s_{l,k}|^2\} = 1$ .

After pulse shaping, the complex baseband transmit waveform of the  $l$ -th user is given by

$$x_l(t) = \sqrt{P_l} \left( \sum_{k=0}^{K-1} s_{l,k} p(t - kT_s) + \xi_l q(t) \right), \quad (1)$$

where  $P_l$  is the average transmit power of user  $l$ ,  $p(t)$  is the pulse-shaping filter,  $T_s$  is the symbol duration,  $\xi_l \in \mathbb{C}$  is a deterministic pilot or bias coefficient, and  $q(t)$  is a known pilot waveform. The deterministic term  $\xi_l q(t)$  is optional and is introduced to allow a nonzero-mean transmitted waveform when required by the subsequent reference-assisted recovery process. If no transmitted pilot or bias is used, we set  $\xi_l = 0$ .

The corresponding real-valued passband signal is expressed as

$$x_l^{\text{RF}}(t) = \sqrt{2} \Re \left\{ x_l(t) e^{j(2\pi f_c t + \varphi_l)} \right\}, \quad l = 1, 2, \dots, L, \quad (2)$$

where  $f_c$  is the carrier frequency and  $\varphi_l$  denotes the initial carrier phase offset of the  $l$ -th user. Let

$$x_l(t) = x_{l,I}(t) + jx_{l,Q}(t), \quad (3)$$

where  $x_{l,I}(t)$  and  $x_{l,Q}(t)$  are the in-phase and quadrature components of the complex baseband waveform, respectively. Then (2) can be equivalently written as

$$x_l^{\text{RF}}(t) = \sqrt{2} [x_{l,I}(t) \cos(2\pi f_c t + \varphi_l) - x_{l,Q}(t) \sin(2\pi f_c t + \varphi_l)]. \quad (4)$$

For compact notation, the multi-user symbol vector at symbol time  $k$  is defined as  $\mathbf{s}_k = [s_{1,k}, s_{2,k}, \dots, s_{L,k}]^T \in \mathcal{S}^L$ , and the multi-user complex baseband waveform vector as  $\mathbf{x}(t) = [x_1(t), x_2(t), \dots, x_L(t)]^T \in \mathbb{C}^L$ . Using (1), the multi-user baseband waveform can be written as

$$\mathbf{x}(t) = \mathbf{D}_P \left( \sum_{k=0}^{K-1} \mathbf{s}_k p(t - kT_s) + \boldsymbol{\xi} q(t) \right), \quad (5)$$

where  $\mathbf{D}_P = \text{diag}(\sqrt{P_1}, \sqrt{P_2}, \dots, \sqrt{P_L})$  is the transmit-power scaling matrix, and  $\boldsymbol{\xi} = [\xi_1, \xi_2, \dots, \xi_L]^T \in \mathbb{C}^L$  collects the deterministic pilot or bias coefficients of all users.

To facilitate the subsequent symbol-level analysis, we further introduce an equivalent matrix representation for the pulse-shaping operation. Consider  $N_t$  baseband observation instants  $\{t_n\}_{n=0}^{N_t-1}$  within the signaling block. For the  $l$ -th user, define the sampled baseband transmit vector as

$$\tilde{\mathbf{x}}_l = [x_l(t_0), x_l(t_1), \dots, x_l(t_{N_t-1})]^T \in \mathbb{C}^{N_t}. \quad (6)$$

From (1), we obtain

$$\tilde{\mathbf{x}}_l = \sqrt{P_l} (\mathbf{P} \mathbf{s}_l + \xi_l \mathbf{q}), \quad (7)$$

where  $\mathbf{P} \in \mathbb{C}^{N_t \times K}$  denotes the pulse-shaping matrix with entries  $[\mathbf{P}]_{n,k} = p(t_n - kT_s)$  for  $n = 0, 1, \dots, N_t - 1$  and  $k = 0, 1, \dots, K - 1$ , and  $\mathbf{q} = [q(t_0), q(t_1), \dots, q(t_{N_t-1})]^T \in \mathbb{C}^{N_t}$  denotes the sampled pilot waveform.

Stacking the sampled baseband transmit vectors of all users gives

$$\tilde{\mathbf{x}} = [\tilde{\mathbf{x}}_1^T, \tilde{\mathbf{x}}_2^T, \dots, \tilde{\mathbf{x}}_L^T]^T \in \mathbb{C}^{LN_t}. \quad (8)$$

Similarly, define the stacked transmitted symbol vector as

$$\mathbf{s} = [\mathbf{s}_1^T, \mathbf{s}_2^T, \dots, \mathbf{s}_L^T]^T \in \mathcal{S}^{LK}. \quad (9)$$

Then the sampled multi-user transmit waveform can be represented as

$$\begin{aligned} \tilde{\mathbf{x}} &= (\mathbf{D}_P \otimes \mathbf{P}) \mathbf{s} + (\mathbf{D}_P \boldsymbol{\xi}) \otimes \mathbf{q}, \\ &= \mathbf{P}_{\text{tx}} \mathbf{s} + \mathbf{x}_0. \end{aligned} \quad (10)$$

where  $\mathbf{P}_{\text{tx}} = \mathbf{D}_P \otimes \mathbf{P} \in \mathbb{C}^{LN_t \times LK}$ ,  $\mathbf{x}_0 = (\mathbf{D}_P \boldsymbol{\xi}) \otimes \mathbf{q} \in \mathbb{C}^{LN_t}$ , and  $\otimes$  denotes the Kronecker product. Thus, the transmitter-side equivalent discrete model is compactly written as

$$\tilde{\mathbf{x}} = \mathbf{P}_{\text{tx}} \mathbf{s} + \mathbf{x}_0. \quad (11)$$

The matrix  $\mathbf{P}_{\text{tx}}$  characterizes the linear mapping from the transmitted constellation symbols to the sampled pulse-shaped waveforms. This representation allows the subsequent receiver model to incorporate pulse shaping, spatial propagation, and holographic RIS observations through an equivalent measurement matrix while keeping the analysis in the discrete symbol domain.

### B. Far-Field Channel Model

As shown in Fig. 1, we consider a holographic RIS receiver composed of a uniform planar array placed on the  $yz$ -plane. The array consists of  $N_y$  elements along the  $y$ -axis and  $N_z$  elements along the  $z$ -axis, resulting in a total of

$$N_r = N_y N_z \quad (12)$$

receiving elements. The position vector of the  $(n, m)$ -th element in the RIS array is defined as

$$\mathbf{r}_{n,m} = [0, y_n, z_m]^T, \quad (13)$$

where  $n = 1, \dots, N_y$  and  $m = 1, \dots, N_z$ , and  $y_n = \left(n - \frac{N_y+1}{2}\right) d_y$ ,  $z_m = \left(m - \frac{N_z+1}{2}\right) d_z$ . Here,  $d_y$  and  $d_z$  denote the inter-element spacings along the  $y$ - and  $z$ -axes, respectively. The centered coordinate system is adopted for notational convenience; using a corner element as the origin leads only to a user-dependent common phase shift, which can be absorbed into the complex channel gain.

Under the far-field assumption, the impinging signal from each user can be modeled as a plane wave across the RIS aperture. Let  $\theta_l$  and  $\phi_l$  denote the angle of arrival (AoA) of the  $l$ -th user, where  $\theta_l$  is the polar angle with respect to the array broadside, i.e., the positive  $x$ -axis, and  $\phi_l$  is the azimuth angle in the  $yz$ -plane measured from the positive  $y$ -axis. The corresponding wave vector is given by

$$\mathbf{k}_l = [k_{l,x}, k_{l,y}, k_{l,z}]^T, \quad (14)$$

with  $k_{l,x} = k_0 \cos \theta_l$ ,  $k_{l,y} = k_0 \sin \theta_l \cos \phi_l$ ,  $k_{l,z} = k_0 \sin \theta_l \sin \phi_l$ ,  $k_0 = \frac{2\pi}{\lambda}$  is the free-space wavenumber and  $\lambda$  is the carrier wavelength.

Following the plane-wave model, the phase of the incident field from user  $l$  at the  $(n, m)$ -th RIS element is determined

by the projection of the wave vector onto the element position, i.e.,

$$\psi_{n,m}^{(l)} = \mathbf{k}_l^T \mathbf{r}_{n,m} = k_{l,y}y_n + k_{l,z}z_m. \quad (15)$$

Since the RIS aperture is assumed to be sufficiently small compared with the propagation distance, the amplitude variation of the incident wave across different RIS elements is neglected for each user. Therefore, the spatial variation across the RIS elements is mainly characterized by the phase profile in (15), while the large-scale path loss, shadowing, and user-dependent phase offset are incorporated into a complex channel gain.

Accordingly, the channel vector from the  $l$ -th user to all RIS receiving elements is modeled as

$$\mathbf{h}_l = \alpha_l \mathbf{v}_l \in \mathbb{C}^{N_r \times 1}, \quad (16)$$

where  $\alpha_l \in \mathbb{C}$  denotes the complex channel gain of user  $l$ , and  $\mathbf{v}_l$  is the normalized array steering vector. The element of  $\mathbf{v}_l$  corresponding to the  $(n, m)$ -th RIS element is given by

$$[\mathbf{v}_l]_{\iota(n,m)} = e^{j(k_{l,y}y_n + k_{l,z}z_m)}, \quad (17)$$

where  $\iota(n, m) = (n-1)N_z + m$  maps the two-dimensional element index  $(n, m)$  to the one-dimensional vector index.

Equivalently, the array steering vector can be written as the Kronecker product of two one-dimensional steering vectors:

$$\mathbf{v}_l = \mathbf{a}_y(\theta_l, \phi_l) \otimes \mathbf{a}_z(\theta_l, \phi_l), \quad (18)$$

where  $\otimes$  denotes the Kronecker product,

$$\mathbf{a}_y(\theta_l, \phi_l) = [e^{jk_{l,y}y_1}, e^{jk_{l,y}y_2}, \dots, e^{jk_{l,y}y_{N_y}}]^T \in \mathbb{C}^{N_y \times 1}, \quad (19)$$

and

$$\mathbf{a}_z(\theta_l, \phi_l) = [e^{jk_{l,z}z_1}, e^{jk_{l,z}z_2}, \dots, e^{jk_{l,z}z_{N_z}}]^T \in \mathbb{C}^{N_z \times 1}. \quad (20)$$

Collecting the channel vectors of all  $L$  users, the multi-user channel matrix between the users and the RIS receiver is expressed as

$$\begin{aligned} \mathbf{H} &= [\mathbf{h}_1, \mathbf{h}_2, \dots, \mathbf{h}_L] \in \mathbb{C}^{N_r \times L} \\ &= [\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_L] \text{diag}(\alpha_1, \alpha_2, \dots, \alpha_L), \end{aligned} \quad (21)$$

where the second line follows from the decomposition of each user channel vector in (16). For compactness, define  $\mathbf{V} = [\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_L] \in \mathbb{C}^{N_r \times L}$  as the RIS array manifold matrix, and  $\mathbf{\Gamma} = \text{diag}(\alpha_1, \alpha_2, \dots, \alpha_L) \in \mathbb{C}^{L \times L}$  as the diagonal matrix of complex channel gains. Then the multi-user channel matrix is compactly written as

$$\mathbf{H} = \mathbf{V}\mathbf{\Gamma}. \quad (22)$$

This decomposition separates the directional spatial response of the RIS, represented by  $\mathbf{V}$ , from the user-dependent complex propagation gains, represented by  $\mathbf{\Gamma}$ . It will be used in the subsequent receiver model to characterize how the pulse-shaped user signals are spatially mapped onto the holographic RIS elements.

### C. Reference-Assisted Holographic Reception Model

At the receiver side, the holographic RIS collects the superimposed signals from all users through its  $N_r$  receiving elements. Using the multi-user channel matrix  $\mathbf{H} \in \mathbb{C}^{N_r \times L}$  defined in the previous subsection, the complex baseband signal impinging on the RIS at time  $t$  is written as

$$\mathbf{r}(t) = \mathbf{H}\mathbf{x}(t). \quad (23)$$

To enable phase information recovery from intensity-only observations, a known reference wave is introduced at the RIS receiver. Let  $\mathbf{b} = [b_1, b_2, \dots, b_{N_r}]^T \in \mathbb{C}^{N_r}$  denote the equivalent complex baseband reference vector, where  $b_i = \rho_i e^{j\varphi_i}$ ,  $i = 1, 2, \dots, N_r$ . Here,  $\rho_i$  and  $\varphi_i$  are the known amplitude and phase of the reference wave injected at the  $i$ -th RIS element, respectively. The corresponding passband reference signal is

$$b_i^{\text{RF}}(t) = \sqrt{2}\Re\{b_i e^{j2\pi f_c t}\} = \sqrt{2}\rho_i \cos(2\pi f_c t + \varphi_i). \quad (24)$$

After superimposing the received signal and the reference wave, the equivalent complex baseband field at the RIS is given by

$$\mathbf{u}(t) = \mathbf{r}(t) + \mathbf{b} = \mathbf{H}\mathbf{x}(t) + \mathbf{b}. \quad (25)$$

Equivalently, the real-valued RF signal at the  $i$ -th RIS element can be expressed as

$$u_i^{\text{RF}}(t) = \sqrt{2}\Re\{u_i(t)e^{j2\pi f_c t}\} + \nu_i^{\text{RF}}(t), \quad (26)$$

where  $u_i(t)$  is the  $i$ -th element of  $\mathbf{u}(t)$  and  $\nu_i^{\text{RF}}(t)$  denotes the RF noise.

The holographic RIS receiver is assumed to employ square-law detection at each element. After low-pass filtering or time averaging over a detection interval  $T_0$ , the intensity measurement at the  $i$ -th element is modeled as

$$z_i(t) = \frac{1}{T_0} \int_t^{t+T_0} |u_i^{\text{RF}}(\tau)|^2 d\tau. \quad (27)$$

Under the narrowband and slowly varying envelope assumptions within the detection interval, the high-frequency components are removed by the low-pass operation, yielding the equivalent baseband intensity observation

$$z_i(t) = |[\mathbf{H}\mathbf{x}(t)]_i + b_i|^2 + w_i(t), \quad i = 1, 2, \dots, N_r, \quad (28)$$

where  $w_i(t)$  denotes the effective post-detection noise and modeling error.

Stacking the observations of all RIS elements gives

$$\mathbf{z}(t) = |\mathbf{H}\mathbf{x}(t) + \mathbf{b}|^2 + \mathbf{w}(t), \quad (29)$$

where  $\mathbf{z}(t) = [z_1(t), z_2(t), \dots, z_{N_r}(t)]^T \in \mathbb{R}^{N_r}$ ,  $\mathbf{w}(t) \in \mathbb{R}^{N_r}$  is the effective intensity-domain noise vector, and the squared modulus in (29) is applied element-wise.

For subsequent symbol-level analysis, we evaluate (29) at the discrete observation instants  $\{t_n\}_{n=0}^{N_t-1}$ . Let

$$\mathbf{x}[n] \triangleq \mathbf{x}(t_n), \quad \mathbf{z}[n] \triangleq \mathbf{z}(t_n), \quad \mathbf{w}[n] \triangleq \mathbf{w}(t_n). \quad (30)$$

Then the discrete-time reference-assisted intensity measurement is

$$\mathbf{z}[n] = |\mathbf{H}\mathbf{x}[n] + \mathbf{b}|^2 + \mathbf{w}[n], \quad n = 0, 1, \dots, N_t - 1. \quad (31)$$

Finally, using the transmitter-side equivalent matrix model in (11), the complete block-level observation can be compactly represented as

$$\mathbf{z} = |\mathbf{G}\mathbf{s} + \mathbf{b}_{\text{eq}}|^2 + \mathbf{w}, \quad (32)$$

where  $\mathbf{s} \in \mathcal{S}^{LK}$  is the stacked transmitted symbol vector,  $\mathbf{G}$  is the equivalent measurement matrix incorporating pulse shaping, propagation, and RIS spatial responses,  $\mathbf{b}_{\text{eq}}$  is the stacked equivalent reference vector, and  $\mathbf{z}$  and  $\mathbf{w}$  collect all intensity observations and post-detection noise samples, respectively. The compact model in (32) will serve as the basis for the subsequent identifiability analysis and detector design.

#### D. Two-Phase Transmission Protocol and Unified Observation Model

Based on the reference-assisted holographic reception model, we consider a two-phase transmission protocol consisting of a training phase and a data transmission phase. In the training phase, the users transmit known pilot waveforms, and the RIS receiver estimates the channel from intensity-only measurements. In the data transmission phase, the users transmit information-bearing symbols, and the receiver detects the transmitted symbols based on the estimated channel. Both phases follow the same reference-assisted intensity observation principle, but the unknown variables are different.

1) *Training Phase:* During the training phase, all users transmit known pilot signals. Let  $\mathbf{x}_p[t] \in \mathbb{C}^L$  denote the known multi-user pilot vector at the  $t$ -th pilot observation, where  $t = 0, 1, \dots, T_p - 1$  and  $T_p$  is the number of pilot observations. The corresponding reference vector is denoted by  $\mathbf{b}_p[t] \in \mathbb{C}^{N_r}$ . According to (31), the received intensity measurement during the training phase for time instants  $t = 0, 1, \dots, T_p - 1$  is given by

$$\mathbf{z}_p[t] = |\mathbf{H}\mathbf{x}_p[t] + \mathbf{b}_p[t]|^2 + \mathbf{w}_p[t]. \quad (33)$$

where  $\mathbf{z}_p[t] \in \mathbb{R}^{N_r}$  is the intensity observation vector, and  $\mathbf{w}_p[t] \in \mathbb{R}^{N_r}$  denotes the effective post-detection noise.

For compact representation, define the vectorized channel as

$$\mathbf{h} = \text{vec}(\mathbf{H}) \in \mathbb{C}^{N_r L}. \quad (34)$$

Using the identity

$$\mathbf{H}\mathbf{x}_p[t] = (\mathbf{x}_p^T[t] \otimes \mathbf{I}_{N_r}) \mathbf{h}, \quad (35)$$

(33) can be rewritten as

$$\mathbf{z}_p[t] = |\mathbf{G}_p[t]\mathbf{h} + \mathbf{b}_p[t]|^2 + \mathbf{w}_p[t], \quad (36)$$

where  $\mathbf{G}_p[t] = \mathbf{x}_p^T[t] \otimes \mathbf{I}_{N_r} \in \mathbb{C}^{N_r \times N_r L}$ . Stacking all pilot observations yields the block-level training model

$$\mathbf{z}_p = |\mathbf{G}_p\mathbf{h} + \mathbf{b}_p|^2 + \mathbf{w}_p, \quad (37)$$

where  $\mathbf{z}_p$ ,  $\mathbf{b}_p$ , and  $\mathbf{w}_p$  collect the corresponding quantities over all pilot observations, and  $\mathbf{G}_p = [\mathbf{G}_p[0], \mathbf{G}_p[1], \dots, \mathbf{G}_p[T_p - 1]]^T$ . If the far-field channel structure in (22) is exploited, the unknown channel vector  $\mathbf{h}$  can be further parameterized by a lower-dimensional channel parameter vector, such as the complex path-gain vector when the angles of arrival are known or pre-estimated.

2) *Data Transmission Phase:* After the training phase, the receiver obtains an estimate of the channel, denoted by  $\hat{\mathbf{H}}$ . In the data transmission phase, the users transmit unknown information-bearing symbols. Let  $\mathbf{s}_d \in \mathcal{S}^{LK_d}$  denote the stacked data-symbol vector within one data block, where  $K_d$  is the number of data symbols per user. Following the transmitter-side equivalent model, the sampled data waveform can be expressed as

$$\tilde{\mathbf{x}}_d = \mathbf{P}_{\text{tx},d}\mathbf{s}_d + \mathbf{x}_{0,d}, \quad (38)$$

where  $\mathbf{P}_{\text{tx},d}$  is the data-phase pulse-shaping matrix and  $\mathbf{x}_{0,d}$  is the deterministic transmit-side pilot or bias component, if present.

Using the estimated channel  $\hat{\mathbf{H}}$ , the block-level data-phase intensity observation can be written as

$$\mathbf{z}_d = |\mathbf{G}_d(\hat{\mathbf{H}})\mathbf{s}_d + \mathbf{b}_{d,\text{eq}}|^2 + \mathbf{w}_d, \quad (39)$$

where  $\mathbf{G}_d(\hat{\mathbf{H}})$  is the equivalent data-phase measurement matrix determined by the estimated channel, the pulse-shaping matrix, and the RIS observation structure. The vector  $\mathbf{b}_{d,\text{eq}}$  denotes the equivalent known offset in the data phase, which includes the receiver-side reference wave and the contribution of any deterministic transmit-side bias. The vector  $\mathbf{w}_d$  represents the effective post-detection noise.

Equations (37) and (39) show that both channel estimation and data detection can be written in the common reference-assisted intensity measurement form

$$\mathbf{z} = |\mathbf{G}\boldsymbol{\theta} + \mathbf{b}_{\text{eq}}|^2 + \mathbf{w}. \quad (40)$$

In the training phase, the unknown variable is the channel-related parameter, e.g.,  $\boldsymbol{\theta} = \mathbf{h}$  or a lower-dimensional parameterization of  $\mathbf{H}$ . In the data transmission phase, the unknown variable is the finite-alphabet symbol vector, i.e.,  $\boldsymbol{\theta} = \mathbf{s}_d$ . Therefore, the two phases share the same phaseless observation structure, while their identifiability conditions and recovery algorithms differ due to the different structures of the unknown variables. The unified model in (40) will serve as the basis for the subsequent identifiability analysis and detector design.

### III. UNIQUENESS AND STABILITY OF REFERENCE-ASSISTED ENERGY DETECTION

This section studies when the reference-assisted square-law receiver has a unique inverse and when that inverse is robust to perturbations. Following the distinction between injectivity and stability in phase retrieval [?], we do not use matrix rank alone as a surrogate for recoverability. Instead, we derive an exact pairwise condition and lower and upper Lipschitz bounds for the nonlinear energy mapping. The analysis applies to both continuous channel estimation and finite-alphabet symbol detection.

#### A. Unified Affine Phase-Retrieval Model

The training- and data-phase observations derived in Section II share the form

$$\mathbf{z} = \mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta}) + \mathbf{w}, \quad \mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta}) \triangleq |\mathbf{G}\boldsymbol{\theta} + \mathbf{b}|^2, \quad (41)$$

where  $\mathbf{G} \in \mathbb{C}^{Q \times D}$ ,  $\boldsymbol{\theta} \in \mathcal{X} \subseteq \mathbb{C}^D$ ,  $\mathbf{b} \in \mathbb{C}^Q$ , and the modulus and square are element-wise. This is an affine phase-retrieval model [9]. In the training phase,  $\boldsymbol{\theta} = \mathbf{h} = \text{vec}(\mathbf{H})$ , or a lower dimensional channel parameter; in the data phase,  $\boldsymbol{\theta} = \mathbf{s}_d \in \mathcal{S}^{LK_d}$ .

The reference term changes the ambiguity class fundamentally. If  $\mathbf{b} = 0$ , then

$$\mathcal{T}_{\mathbf{G},0}(e^{j\varphi}\boldsymbol{\theta}) = \mathcal{T}_{\mathbf{G},0}(\boldsymbol{\theta}), \quad (42)$$

and uniqueness can only be defined modulo a global phase. If  $\mathbf{b} \neq 0$ , the expansion

$$\mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta}) = |\mathbf{G}\boldsymbol{\theta}|^2 + |\mathbf{b}|^2 + 2\text{Re}\{\mathbf{b}^* \odot \mathbf{G}\boldsymbol{\theta}\} \quad (43)$$

contains phase-sensitive interference, so absolute uniqueness on  $\mathcal{X}$  becomes possible. A nonzero reference is not sufficient by itself, however: its phases must probe every feasible field difference in nondegenerate real directions.

To quantify this statement, define the realification

$$\boldsymbol{\theta}_{\mathbb{R}} \triangleq [\text{Re}\{\boldsymbol{\theta}\}^T \quad \text{Im}\{\boldsymbol{\theta}\}^T]^T \in \mathbb{R}^{2D} \quad (44)$$

and, for  $\boldsymbol{\theta}_1 \neq \boldsymbol{\theta}_2$ , the energy-domain distance

$$d_E(\boldsymbol{\theta}_1, \boldsymbol{\theta}_2) \triangleq \|\mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta}_1) - \mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta}_2)\|_2. \quad (45)$$

The optimal lower and upper stability bounds on  $\mathcal{X}$  are

$$\alpha_{\mathcal{X}} \triangleq \inf_{\substack{\boldsymbol{\theta}_1, \boldsymbol{\theta}_2 \in \mathcal{X} \\ \boldsymbol{\theta}_1 \neq \boldsymbol{\theta}_2}} \frac{d_E(\boldsymbol{\theta}_1, \boldsymbol{\theta}_2)}{\|\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2\|_2}, \quad (46)$$

$$\beta_{\mathcal{X}} \triangleq \sup_{\substack{\boldsymbol{\theta}_1, \boldsymbol{\theta}_2 \in \mathcal{X} \\ \boldsymbol{\theta}_1 \neq \boldsymbol{\theta}_2}} \frac{d_E(\boldsymbol{\theta}_1, \boldsymbol{\theta}_2)}{\|\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2\|_2}. \quad (47)$$

Thus, the inverse problem is globally stable on  $\mathcal{X}$  if  $0 < \alpha_{\mathcal{X}} \leq \beta_{\mathcal{X}} < \infty$ . The lower bound is the important one: it excludes pairs that are far apart in parameter space but arbitrarily close in energy space.

### B. Exact Secant Identity and Global Uniqueness

The quadratic structure of (41) gives an exact identity, which is stronger than a first-order approximation. For a point  $\boldsymbol{\vartheta} \in \mathbb{C}^D$ , define the real Jacobian

$$\mathbf{J}(\boldsymbol{\vartheta}) \triangleq 2[\text{Re}\{\text{diag}[(\mathbf{G}\boldsymbol{\vartheta} + \mathbf{b})^*]\mathbf{G}\}, \\ -\text{Im}\{\text{diag}[(\mathbf{G}\boldsymbol{\vartheta} + \mathbf{b})^*]\mathbf{G}\}] \in \mathbb{R}^{Q \times 2D}. \quad (48)$$

**Lemma 1** (Exact midpoint factorization). *For any  $\boldsymbol{\theta}_1, \boldsymbol{\theta}_2 \in \mathbb{C}^D$ , let  $\boldsymbol{\mu} = (\boldsymbol{\theta}_1 + \boldsymbol{\theta}_2)/2$  and  $\boldsymbol{\delta} = \boldsymbol{\theta}_1 - \boldsymbol{\theta}_2$ . Then*

$$\mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta}_1) - \mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta}_2) = \mathbf{J}(\boldsymbol{\mu})\boldsymbol{\delta}_{\mathbb{R}}. \quad (49)$$

*Proof.* Applying  $|a|^2 - |c|^2 = \text{Re}\{(a+c)^*(a-c)\}$  element-wise gives

$$\text{Re}\{[\mathbf{G}(\boldsymbol{\theta}_1 + \boldsymbol{\theta}_2) + 2\mathbf{b}]^* \odot \mathbf{G}(\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2)\}, \quad (50)$$

which equals the right-hand side of (49) after realification.  $\square$

**Theorem 1** (Pairwise uniqueness criterion). *The mapping  $\mathcal{T}_{\mathbf{G},\mathbf{b}}$  is injective on  $\mathcal{X}$  if and only if, for every distinct  $\boldsymbol{\theta}_1, \boldsymbol{\theta}_2 \in \mathcal{X}$ ,*

$$\mathbf{J}\left(\frac{\boldsymbol{\theta}_1 + \boldsymbol{\theta}_2}{2}\right)(\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2)_{\mathbb{R}} \neq \mathbf{0}. \quad (51)$$

*A necessary condition, independent of the reference, is*

$$\begin{aligned} \text{Null}(\mathbf{G}) \cap \mathcal{D}_{\mathcal{X}} &= \emptyset, \\ \mathcal{D}_{\mathcal{X}} &\triangleq \{\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2 : \boldsymbol{\theta}_1, \boldsymbol{\theta}_2 \in \mathcal{X}, \boldsymbol{\theta}_1 \neq \boldsymbol{\theta}_2\}. \end{aligned} \quad (52)$$

*Proof.* The first claim follows directly from Lemma 1. For the second, if  $\mathbf{G}(\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2) = \mathbf{0}$ , then the two complex fields, and hence their intensities for every  $\mathbf{b}$ , are identical.  $\square$

Theorem 1 makes clear why  $\text{rank}(\mathbf{G}) = D$  is necessary for a full-dimensional continuous parameter set but is not sufficient for phase retrieval: even when  $\mathbf{G}\boldsymbol{\delta} \neq \mathbf{0}$ , its real projection determined jointly by the midpoint field and the reference may vanish. For a finite constellation, full column rank is not necessary; only the finite difference set in (52) must avoid the null space.

Let

$$\mathcal{M}_{\mathcal{X}} \triangleq \{(\boldsymbol{\theta}_1 + \boldsymbol{\theta}_2)/2 : \boldsymbol{\theta}_1, \boldsymbol{\theta}_2 \in \mathcal{X}\} \quad (53)$$

be the midpoint set. The exact identity immediately yields computable two-sided bounds.

**Proposition 1** (Jacobian stability bounds). *Define*

$$\underline{\sigma}_J \triangleq \inf_{\boldsymbol{\mu} \in \mathcal{M}_{\mathcal{X}}} \sigma_{\min}(\mathbf{J}(\boldsymbol{\mu})), \quad \bar{\sigma}_J \triangleq \sup_{\boldsymbol{\mu} \in \mathcal{M}_{\mathcal{X}}} \sigma_{\max}(\mathbf{J}(\boldsymbol{\mu})). \quad (54)$$

*Then*

$$\underline{\sigma}_J \|\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2\|_2 \leq d_E(\boldsymbol{\theta}_1, \boldsymbol{\theta}_2) \leq \bar{\sigma}_J \|\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2\|_2 \quad (55)$$

*for all pairs in  $\mathcal{X}$ . Consequently,  $\alpha_{\mathcal{X}} \geq \underline{\sigma}_J$  and  $\beta_{\mathcal{X}} \leq \bar{\sigma}_J$ . If  $\|\boldsymbol{\theta}\|_2 \leq R_{\mathcal{X}}$  on  $\mathcal{X}$ , then*

$$\beta_{\mathcal{X}} \leq 2\|\mathbf{G}\|_2 (\|\mathbf{G}\|_2 R_{\mathcal{X}} + \|\mathbf{b}\|_{\infty}). \quad (56)$$

*Proof.* The first statement follows by applying the singular-value inequalities to (49). For the explicit upper bound, use  $\|\text{diag}(\mathbf{u}^*)\mathbf{G}\|_2 \leq \|\mathbf{u}\|_{\infty}\|\mathbf{G}\|_2$  and  $\|\mathbf{G}\boldsymbol{\mu} + \mathbf{b}\|_{\infty} \leq \|\mathbf{G}\|_2 R_{\mathcal{X}} + \|\mathbf{b}\|_{\infty}$ .  $\square$

The bound  $\underline{\sigma}_J > 0$  is a convenient sufficient condition for global stability, but it can be conservative because it tests every direction at every midpoint, including directions that may not connect two feasible points. The exact constant in (46), or equivalently the restricted secants in (49), is the sharp criterion. Notice also that injectivity alone need not give a useful uniform lower bound on a continuous set; this is precisely the distinction between uniqueness and stability emphasized in [?].

### C. Continuous Channel Identifiability

For a continuous channel parameter, local identifiability at  $\boldsymbol{\theta}$  requires

$$\text{rank}(\mathbf{J}(\boldsymbol{\theta})) = 2D, \quad Q \geq 2D. \quad (57)$$

The smallest singular value determines the local inverse sensitivity:

$$\liminf_{\|\boldsymbol{\delta}\|_2 \rightarrow 0} \frac{d_E(\boldsymbol{\theta} + \boldsymbol{\delta}, \boldsymbol{\theta})}{\|\boldsymbol{\delta}\|_2} = \sigma_{\min}(\mathbf{J}(\boldsymbol{\theta})). \quad (58)$$

Local full rank does not rule out a distant ambiguous channel. Global uniqueness still requires (51) for all pairs, while a positive  $\alpha_{\mathcal{X}}$  rules out both exact and near-ambiguities.

These constants also have a direct error interpretation. Let  $\theta_* \in \mathcal{X}$  generate (41), and let  $\hat{\theta}$  be a global minimizer of the intensity least-squares criterion. If  $\alpha_{\mathcal{X}} > 0$ , then

$$\|\hat{\theta} - \theta_*\|_2 \leq \frac{2\|\mathbf{w}\|_2}{\alpha_{\mathcal{X}}}. \quad (59)$$

Indeed, optimality gives a residual no larger than  $\|\mathbf{w}\|_2$ ; the triangle inequality followed by the lower Lipschitz bound gives (59).

For full-channel training,

$$\mathbf{G}_p = \mathbf{X}_p \otimes \mathbf{I}_{N_r}, \quad \mathbf{X}_p \in \mathbb{C}^{T_p \times L}, \quad (60)$$

and therefore

$$\begin{aligned} \text{rank}(\mathbf{G}_p) &= N_r \text{rank}(\mathbf{X}_p), \\ \sigma_{\min}(\mathbf{G}_p) &= \sigma_{\min}(\mathbf{X}_p), \\ \kappa(\mathbf{G}_p) &= \kappa(\mathbf{X}_p). \end{aligned} \quad (61)$$

Thus,  $\text{rank}(\mathbf{X}_p) = L$  and  $T_p \geq L$  are necessary to preserve the complex channel field. They are not sufficient for the intensity mapping. With one real intensity per RIS element and pilot time,  $Q = N_r T_p$  and  $D = N_r L$ , so the dimension condition in (57) strengthens the counting requirement to

$$T_p \geq 2L. \quad (62)$$

Even when this count is met, the pilot and reference phases must make  $\mathbf{J}_p$  full column rank and well conditioned.

If the far-field model  $\mathbf{H} = \mathbf{V}\mathbf{\Gamma}$  is used and the angles are known, the unknown can be reduced to  $\alpha = [\alpha_1, \dots, \alpha_L]^T$ , with

$$\mathbf{G}_\alpha = \begin{bmatrix} \mathbf{V} \text{diag}(\mathbf{x}_p[0]) \\ \vdots \\ \mathbf{V} \text{diag}(\mathbf{x}_p[T_p - 1]) \end{bmatrix}. \quad (63)$$

The necessary count becomes  $N_r T_p \geq 2L$ . In addition,  $\mathbf{G}_\alpha$  must preserve every gain direction and the reference must provide two independent real projections of those directions. Closely spaced angles reduce  $\sigma_{\min}(\mathbf{V})$  and hence both the field-domain and energy-domain stability margins.

#### D. Finite-Alphabet Uniqueness and Noise Margin

For data detection, let  $\mathcal{X}_d = \mathcal{S}^{LK_d}$ . Since this set is finite, global uniqueness and a positive minimum separation are equivalent. Define

$$d_{\min,d}^{(E)} \triangleq \min_{\substack{\mathbf{s}_1, \mathbf{s}_2 \in \mathcal{X}_d \\ \mathbf{s}_1 \neq \mathbf{s}_2}} \left\| |\mathbf{G}_d \mathbf{s}_1 + \mathbf{b}_d|^2 - |\mathbf{G}_d \mathbf{s}_2 + \mathbf{b}_d|^2 \right\|_2. \quad (64)$$

**Corollary 1** (Discrete uniqueness and stability). *The symbols are uniquely identifiable if and only if  $d_{\min,d}^{(E)} > 0$ . Equivalently, for every distinct pair,*

$$\mathbf{J}_d \left( \frac{\mathbf{s}_1 + \mathbf{s}_2}{2} \right) (\mathbf{s}_1 - \mathbf{s}_2)_{\mathbb{R}} \neq \mathbf{0}. \quad (65)$$

Moreover, the nearest-intensity detector is guaranteed to return the transmitted symbol vector whenever

$$\|\mathbf{w}\|_2 < \frac{d_{\min,d}^{(E)}}{2}. \quad (66)$$

For additive white Gaussian post-detection noise with covariance  $\sigma_w^2 \mathbf{I}$ , the pairwise ML error probability is

$$P(\mathbf{s}_1 \rightarrow \mathbf{s}_2) = Q\left(\frac{d_E(\mathbf{s}_1, \mathbf{s}_2)}{2\sigma_w}\right), \quad (67)$$

so (64) is both a uniqueness certificate and the worst-case noise margin.

The data-phase field matrix has the factorization

$$\mathbf{G}_d = (\mathbf{I}_{N_t} \otimes \hat{\mathbf{H}}) \mathbf{\Pi} \mathbf{P}_{\text{tx},d}. \quad (68)$$

When the factors have full column rank,

$$\sigma_{\min}(\mathbf{G}_d) \geq \sigma_{\min}(\hat{\mathbf{H}}) \sigma_{\min}(\mathbf{P}_{\text{tx},d}), \quad (69)$$

$$\sigma_{\min}(\hat{\mathbf{H}}) \approx \sigma_{\min}(\mathbf{V}\mathbf{\Gamma}) \geq \sigma_{\min}(\mathbf{V}) \min_l |\alpha_l|. \quad (70)$$

These inequalities quantify only preservation in the complex field domain. Theorem 1 adds the missing condition: the reference-induced real projection must also be nonzero for every constellation secant.

#### E. A Constructive Four-Phase Reference Bound

The preceding conditions are exact but depend on the unknown midpoint. A phase-complete reference pattern gives a simple midpoint-independent sufficient condition. Suppose a base field sample  $u_q = [\mathbf{G}\theta]_q$  is repeated with four known references

$$b_{q,k} = \rho_q e^{j\phi_k}, \quad \phi_k \in \{0, \pi/2, \pi, 3\pi/2\}, \quad \rho_q > 0. \quad (71)$$

Denote the resulting stacked mapping by  $\mathcal{T}_{4\text{ph}}(\theta)$ .

**Theorem 2** (Four-phase global lower bound). *For any  $\theta_1, \theta_2$ ,*

$$\begin{aligned} \|\mathcal{T}_{4\text{ph}}(\theta_1) - \mathcal{T}_{4\text{ph}}(\theta_2)\|_2^2 \\ = 4\| |\mathbf{G}\theta_1|^2 - |\mathbf{G}\theta_2|^2 \|_2^2 + 8 \sum_q \rho_q^2 |[\mathbf{G}(\theta_1 - \theta_2)]_q|^2. \end{aligned} \quad (72)$$

Consequently, with  $\rho_{\min} = \min_q \rho_q$ ,

$$\|\mathcal{T}_{4\text{ph}}(\theta_1) - \mathcal{T}_{4\text{ph}}(\theta_2)\|_2 \geq 2\sqrt{2}\rho_{\min} \|\mathbf{G}(\theta_1 - \theta_2)\|_2. \quad (73)$$

If  $\mathbf{G}$  has full column rank, then

$$\alpha_{\mathcal{X}} \geq 2\sqrt{2}\rho_{\min} \sigma_{\min}(\mathbf{G}) > 0 \quad (74)$$

for every feasible set  $\mathcal{X}$ . For a finite alphabet, full column rank can be replaced by  $\text{Null}(\mathbf{G}) \cap \mathcal{D}_{\mathcal{X}} = \emptyset$ .

*Proof.* For one field coordinate, put  $c_q = |u_{1,q}|^2 - |u_{2,q}|^2$  and  $\Delta u_q = u_{1,q} - u_{2,q}$ . Summing  $[c_q + 2\rho_q \text{Re}\{e^{-j\phi_k} \Delta u_q\}]^2$  over the four phases cancels all cross terms and gives  $4c_q^2 + 8\rho_q^2 |\Delta u_q|^2$ . Summing over  $q$  proves (72); the remaining claims follow from singular value or restricted-difference bounds.  $\square$

The construction also has a transparent holographic interpretation. Opposite reference phases cancel the unknown self-intensity:

$$\text{Re}\{u_q\} = \frac{z_{q,0} - z_{q,\pi}}{4\rho_q}, \quad \text{Im}\{u_q\} = \frac{z_{q,\pi/2} - z_{q,3\pi/2}}{4\rho_q}. \quad (75)$$

Thus, four phase-shifted energy measurements recover two orthogonal real projections of each complex field sample. This

proves explicitly that reference phase diversity can remove the global phase ambiguity and yields the stability lower bound in (74). Fewer or hardware-constrained phases may still be unique, but their quality should be evaluated through  $\alpha_{\mathcal{X}}$ ,  $\underline{\sigma}_{\mathcal{J}}$ , or  $d_{\min, d}^{(E)}$ , rather than through the rank of  $\mathbf{G}$  alone.

The resulting design criteria are therefore

$$\max_{\mathbf{b}_p \in \mathcal{B}_p} \inf_{\boldsymbol{\theta} \in \mathcal{X}_p} \sigma_{\min}[\mathbf{J}_p(\boldsymbol{\theta})] \quad \text{and} \quad \max_{\mathbf{b}_d \in \mathcal{B}_d} d_{\min, d}^{(E)} \quad (76)$$

for continuous training and discrete data detection, respectively. The first maximizes a uniform local/global lower bound, whereas the second directly maximizes the finite-alphabet decision margin.

#### IV. ESTIMATION AND DETECTION ALGORITHMS

Based on the uniqueness and stability analysis in Section III, this section develops practical recovery algorithms for the proposed holographic RIS receiver. The key point is that the algorithms should not be chosen only according to the algebraic rank of the equivalent matrix. Instead, the training stage should operate in a region where the intensity-domain Jacobian is well conditioned, while the data detector should preserve a positive energy-domain distance between finite-alphabet candidates.

The resulting receiver follows a two-stage structure. In the training stage, the unknown variables are the DOAs and the complex path gains. They are continuous parameters, and the far-field model provides a differentiable low-dimensional representation. A damped Gauss–Newton refinement is therefore used after a coarse initialization. In the data stage, the channel estimate is fixed and the unknown vector belongs to a finite constellation. Maximum-likelihood (ML) detection gives the finite-alphabet benchmark associated with the energy-domain minimum distance, whereas a reference-assisted Gerchberg–Saxton (GS) detector provides a lower-complexity iterative implementation for repeated data blocks.

##### A. Overall Two-Stage Recovery Framework

The proposed receiver operates in two stages: a training stage for joint DOA and channel-gain estimation and a data stage for symbol detection.

During the training stage, the users transmit known pilot vectors  $\mathbf{x}_p[t] \in \mathbb{C}^L$ ,  $t = 0, 1, \dots, T_p - 1$ . The far-field structure in (22) is reused here by writing

$$\mathbf{H}(\boldsymbol{\eta}) = \mathbf{V}(\boldsymbol{\theta}, \boldsymbol{\varphi}) \mathbf{\Gamma}(\boldsymbol{\alpha}), \quad (77)$$

where  $\mathbf{V}(\boldsymbol{\theta}, \boldsymbol{\varphi})$  is the RIS array manifold determined by the users' DOAs, and  $\mathbf{\Gamma}(\boldsymbol{\alpha}) = \text{diag}(\alpha_1, \dots, \alpha_L)$  collects the complex path gains. Thus, the training algorithm only needs to estimate the physical parameters that determine the manifold and gains, rather than all entries of  $\mathbf{H}$  independently. We collect these unknowns in the real-valued vector

$$\boldsymbol{\eta} = [\boldsymbol{\theta}^T \quad \boldsymbol{\varphi}^T \quad \text{Re}\{\boldsymbol{\alpha}\}^T \quad \text{Im}\{\boldsymbol{\alpha}\}^T]^T \in \mathbb{R}^{D_\eta}, \quad (78)$$

where  $\boldsymbol{\theta}$  and  $\boldsymbol{\varphi}$  represent the two angular coordinates, while  $\text{Re}\{\boldsymbol{\alpha}\}$  and  $\text{Im}\{\boldsymbol{\alpha}\}$  represent the complex gains in real form; hence  $D_\eta = 4L$  in the two-dimensional DOA case. If

only one angular parameter is considered for each user, the corresponding angular block is removed and the parameter dimension is reduced accordingly.

Define the stacked noiseless received field during the training stage as

$$\mathbf{g}_p(\boldsymbol{\eta}) = \begin{bmatrix} \mathbf{H}(\boldsymbol{\eta})\mathbf{x}_p[0] \\ \mathbf{H}(\boldsymbol{\eta})\mathbf{x}_p[1] \\ \vdots \\ \mathbf{H}(\boldsymbol{\eta})\mathbf{x}_p[T_p - 1] \end{bmatrix} \in \mathbb{C}^{Q_p}, \quad (79)$$

where  $Q_p = N_r T_p$  is the total number of training-stage intensity observations. The stacked training model is

$$\mathbf{z}_p = |\mathbf{g}_p(\boldsymbol{\eta}) + \mathbf{b}_p|^2 + \mathbf{w}_p, \quad (80)$$

where  $\mathbf{b}_p \in \mathbb{C}^{Q_p}$  is the stacked known reference vector.

After estimating  $\boldsymbol{\eta}$ , the channel estimate is reconstructed as

$$\hat{\mathbf{H}} = \mathbf{H}(\hat{\boldsymbol{\eta}}). \quad (81)$$

During the data stage, the estimated channel is incorporated into the equivalent data-phase measurement matrix

$$\hat{\mathbf{G}}_d = \mathbf{G}_d(\hat{\mathbf{H}}). \quad (82)$$

The data observation is then modeled as

$$\mathbf{z}_d = |\hat{\mathbf{G}}_d \mathbf{s}_d + \mathbf{b}_d|^2 + \mathbf{w}_d, \quad (83)$$

where  $\mathbf{s}_d \in \mathcal{S}^{D_s}$  is the unknown finite-alphabet symbol vector and  $D_s = LK_d$ .

The stability results in Section III lead to three algorithmic design rules. First, the pilot matrix and RIS observation diversity must make the field response sensitive to different channel parameters; otherwise no nonlinear solver can restore the lost information. Second, the training reference should make the real Jacobian  $\mathbf{J}_p(\boldsymbol{\eta})$  well conditioned over the feasible angular region. Third, the data-phase reference should maximize the estimated minimum energy-domain distance between constellation vectors, since this distance is the decision margin of the ML detector and a useful indicator for the iterative detectors. The overall receiver is summarized in Algorithm 1.

The two stages are therefore connected through the estimated channel but employ different recovery mechanisms. The Gauss–Newton estimator exploits the differentiable low-dimensional channel parameterization, whereas the reference-assisted Gerchberg–Saxton detector exploits repeated projections and the finite-alphabet structure of communication symbols.

##### B. Joint DOA and Channel-Gain Estimation

1) *Nonlinear Least-Squares Formulation:* Define the predicted training-stage complex field as

$$\mathbf{u}_p(\boldsymbol{\eta}) = \mathbf{g}_p(\boldsymbol{\eta}) + \mathbf{b}_p. \quad (84)$$

The corresponding predicted intensity is

$$\hat{\mathbf{z}}_p(\boldsymbol{\eta}) = |\mathbf{u}_p(\boldsymbol{\eta})|^2. \quad (85)$$

The joint DOA and channel-gain estimation problem is formulated as

$$\hat{\boldsymbol{\eta}} = \arg \min_{\boldsymbol{\eta} \in \Omega} F_p(\boldsymbol{\eta}), \quad (86)$$



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**Algorithm 1** Two-Stage Reference-Assisted Holographic Reception
 

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**Require:** Pilot vectors  $\mathbf{x}_p[t]$ , training observations  $\mathbf{z}_p$ , data observations  $\mathbf{z}_d$ , RIS geometry, reference vectors  $\mathbf{b}_p$  and  $\mathbf{b}_d$ , and constellation set  $\mathcal{S}$ .

**Ensure:** Estimated DOAs, estimated channel  $\hat{\mathbf{H}}$ , and detected symbols  $\hat{\mathbf{s}}_d$ .

- 1: **Training stage:**
  - 2: Construct the parametric field model  $\mathbf{g}_p(\boldsymbol{\eta})$  from the pilot vectors and RIS array manifold.
  - 3: Obtain a coarse initial estimate  $\boldsymbol{\eta}^{(0)}$ .
  - 4: Refine the DOAs and complex channel gains using the damped Gauss–Newton algorithm.
  - 5: Reconstruct  $\hat{\mathbf{H}} = \mathbf{H}(\hat{\boldsymbol{\eta}})$ .
  - 6: **Data stage:**
  - 7: Construct  $\hat{\mathbf{G}}_d = \mathbf{G}_d(\hat{\mathbf{H}})$ .
  - 8: Detect  $\mathbf{s}_d$  using maximum-likelihood detection or the proposed reference-assisted Gerchberg–Saxton detector.
  - 9: Output the estimated DOAs,  $\hat{\mathbf{H}}$ , and  $\hat{\mathbf{s}}_d$ .
- 

where

$$F_p(\boldsymbol{\eta}) = \frac{1}{2} \left\| \mathbf{z}_p - |\mathbf{g}_p(\boldsymbol{\eta}) + \mathbf{b}_p|^2 \right\|_2^2, \quad (87)$$

and  $\Omega$  denotes the feasible parameter region. For example, the angular parameters can be restricted according to

$$0 \leq \theta_l \leq \frac{\pi}{2}, \quad -\pi \leq \varphi_l < \pi. \quad (88)$$

This formulation directly exploits the far-field array structure. Compared with estimating all entries of  $\mathbf{H} \in \mathbb{C}^{N_r \times L}$  independently, the parametric model substantially reduces the number of unknowns when the number of RIS elements is large. More importantly, the optimization variable is exactly the physical quantity that determines the array manifold. Hence, the Jacobian used by the numerical solver also has a direct identifiability interpretation: poorly separated DOAs or weak path gains appear as small singular values of the intensity-domain Jacobian.

2) *Jacobian of the Parametric Intensity Mapping:* At the  $t$ -th pilot instant, the noiseless field is

$$\mathbf{g}_p[t] = \sum_{l=1}^L \alpha_l x_{p,l}[t] \mathbf{v}(\theta_l, \varphi_l), \quad (89)$$

where  $x_{p,l}[t]$  is the pilot symbol transmitted by user  $l$ .

For the  $(n, m)$ -th RIS element, the phase of the  $l$ -th steering vector is

$$\psi_{n,m}^{(l)} = k_0 \sin \theta_l (y_n \cos \varphi_l + z_m \sin \varphi_l). \quad (90)$$

Hence,

$$[\mathbf{v}(\theta_l, \varphi_l)]_{l(n,m)} = e^{j\psi_{n,m}^{(l)}}. \quad (91)$$

Define the real-valued derivative vectors  $\mathbf{d}_{\theta,l} \in \mathbb{R}^{N_r}$  and  $\mathbf{d}_{\varphi,l} \in \mathbb{R}^{N_r}$ , whose entries are

$$[\mathbf{d}_{\theta,l}]_{l(n,m)} = k_0 \cos \theta_l (y_n \cos \varphi_l + z_m \sin \varphi_l), \quad (92a)$$

$$[\mathbf{d}_{\varphi,l}]_{l(n,m)} = k_0 \sin \theta_l (-y_n \sin \varphi_l + z_m \cos \varphi_l). \quad (92b)$$

The steering-vector derivatives are therefore

$$\frac{\partial \mathbf{v}(\theta_l, \varphi_l)}{\partial \theta_l} = j \mathbf{d}_{\theta,l} \odot \mathbf{v}(\theta_l, \varphi_l), \quad (93a)$$

$$\frac{\partial \mathbf{v}(\theta_l, \varphi_l)}{\partial \varphi_l} = j \mathbf{d}_{\varphi,l} \odot \mathbf{v}(\theta_l, \varphi_l). \quad (93b)$$

Using (89), and writing  $\mathbf{v}_l = \mathbf{v}(\theta_l, \varphi_l)$ ,  $\alpha_{l,R} = \text{Re}\{\alpha_l\}$ , and  $\alpha_{l,I} = \text{Im}\{\alpha_l\}$ , the field derivatives at pilot instant  $t$  are

$$\frac{\partial \mathbf{g}_p[t]}{\partial \theta_l} = \alpha_l x_{p,l}[t] \frac{\partial \mathbf{v}_l}{\partial \theta_l}, \quad (94a)$$

$$\frac{\partial \mathbf{g}_p[t]}{\partial \varphi_l} = \alpha_l x_{p,l}[t] \frac{\partial \mathbf{v}_l}{\partial \varphi_l}, \quad (94b)$$

$$\frac{\partial \mathbf{g}_p[t]}{\partial \alpha_{l,R}} = x_{p,l}[t] \mathbf{v}_l, \quad (94c)$$

$$\frac{\partial \mathbf{g}_p[t]}{\partial \alpha_{l,I}} = j x_{p,l}[t] \mathbf{v}_l. \quad (94d)$$

The derivatives over all pilot instants are stacked in the same order as  $\mathbf{g}_p(\boldsymbol{\eta})$ . For the  $r$ -th real parameter  $\eta_r$ , the corresponding column of the intensity-domain Jacobian is

$$[\mathbf{J}_p(\boldsymbol{\eta})]_{:,r} = 2 \text{Re} \left\{ \mathbf{u}_p^*(\boldsymbol{\eta}) \odot \frac{\partial \mathbf{g}_p(\boldsymbol{\eta})}{\partial \eta_r} \right\}. \quad (95)$$

Equivalently,

$$\mathbf{J}_p(\boldsymbol{\eta}) = 2 \text{Re} \left\{ \text{diag}(\mathbf{u}_p^*(\boldsymbol{\eta})) \frac{\partial \mathbf{g}_p(\boldsymbol{\eta})}{\partial \boldsymbol{\eta}^\top} \right\} \in \mathbb{R}^{Q_p \times D_\eta}. \quad (96)$$

The same Jacobian determines both the local identifiability discussed in Section III and the numerical behavior of the Gauss–Newton iterations. In particular, a small  $\sigma_{\min}(\mathbf{J}_p)$  indicates that certain perturbations in the DOAs or path gains produce only weak intensity variations, leading to both poor estimation stability and poorly conditioned Gauss–Newton updates. This is the reason for using phase-diverse references during training: the reference term changes the real projection directions in the Jacobian and can convert a nearly invisible complex-field perturbation into a measurable intensity perturbation.

3) *Damped Gauss–Newton Update:* At iteration  $i$ , define

$$\mathbf{u}_p^{(i)} = \mathbf{g}_p(\boldsymbol{\eta}^{(i)}) + \mathbf{b}_p, \quad (97)$$

and

$$\mathbf{e}_p^{(i)} = \mathbf{z}_p - \left| \mathbf{u}_p^{(i)} \right|^2. \quad (98)$$

The residual is locally approximated as

$$\mathbf{e}_p(\boldsymbol{\eta}^{(i)} + \Delta \boldsymbol{\eta}) \approx \mathbf{e}_p^{(i)} - \mathbf{J}_p^{(i)} \Delta \boldsymbol{\eta}, \quad (99)$$

where  $\mathbf{J}_p^{(i)} = \mathbf{J}_p(\boldsymbol{\eta}^{(i)})$  denotes the intensity-domain Jacobian evaluated at the current iterate.

The damped Gauss–Newton increment is obtained from

$$\Delta \boldsymbol{\eta}^{(i)} = \left[ \left( \mathbf{J}_p^{(i)} \right)^\top \mathbf{J}_p^{(i)} + \lambda^{(i)} \mathbf{I} \right]^{-1} \left( \mathbf{J}_p^{(i)} \right)^\top \mathbf{e}_p^{(i)}, \quad (100)$$

where  $\lambda^{(i)} \geq 0$  is a damping parameter. The candidate parameter vector is

$$\boldsymbol{\eta}_{\text{trial}}^{(i+1)} = \Pi_{\Omega} \left( \boldsymbol{\eta}^{(i)} + \Delta \boldsymbol{\eta}^{(i)} \right), \quad (101)$$

where  $\Pi_{\Omega}(\cdot)$  projects the angular parameters onto the feasible region.

An adaptive damping strategy can be employed. If

$$F_{\text{p}} \left( \boldsymbol{\eta}_{\text{trial}}^{(i+1)} \right) < F_{\text{p}} \left( \boldsymbol{\eta}^{(i)} \right), \quad (102)$$

the trial point is accepted and the damping parameter is reduced. Otherwise, the trial point is rejected and the damping parameter is increased. This procedure provides greater robustness than an undamped Gauss–Newton iteration when the initialization is not sufficiently close to the optimum or when the Jacobian is poorly conditioned. In implementation, one may use

$$\lambda^{(i+1)} = \begin{cases} \max\{\lambda_{\min}, \lambda^{(i)}/\tau\}, & \text{accepted,} \\ \min\{\lambda_{\max}, \tau\lambda^{(i)}\}, & \text{rejected,} \end{cases} \quad (103)$$

where  $\tau > 1$ . The damping term prevents the update from moving along weakly observable directions when  $\mathbf{J}_{\text{p}}^{(i)}$  has small singular values.

4) *Initialization*: Since the nonlinear least-squares problem in (86) is nonconvex, a suitable initialization is important. The initialization should be consistent with the uniqueness analysis in Section III: it should first recover, or approximately recover, the complex field observed at the RIS, and only then fit the far-field angular model.

If the four-phase reference pattern in Theorem 2 is used during training, the complex field can be initialized directly from opposite reference phases. For a fixed pilot time and RIS element, let  $z_{q,0}$ ,  $z_{q,\pi/2}$ ,  $z_{q,\pi}$ , and  $z_{q,3\pi/2}$  denote the four intensity measurements obtained with reference magnitude  $\rho_q$ . The field sample  $u_q = [\mathbf{g}_{\text{p}}]_q$  admits the estimate

$$\hat{u}_q = \frac{z_{q,0} - z_{q,\pi}}{4\rho_q} + j \frac{z_{q,\pi/2} - z_{q,3\pi/2}}{4\rho_q}. \quad (104)$$

Stacking all pilot times gives a complex field estimate  $\hat{\mathbf{g}}_{\text{p}}$ , from which an unstructured channel estimate can be obtained by least squares using

$$\hat{\mathbf{g}}_{\text{p}} \approx \mathbf{G}_{\text{p}} \mathbf{h}, \quad \mathbf{h} = \text{vec}(\mathbf{H}). \quad (105)$$

When four separate reference phases are not available, a practical initial estimate can be obtained from the reference-dominated linear approximation.

Let  $\mathbf{h} = \text{vec}(\mathbf{H})$  and let  $\mathbf{G}_{\text{p}}$  denote the training matrix satisfying  $\mathbf{g}_{\text{p}} = \mathbf{G}_{\text{p}} \mathbf{h}$ . When the reference field is sufficiently strong, the self-intensity term  $|\mathbf{G}_{\text{p}} \mathbf{h}|^2$  can be neglected for initialization, giving

$$\mathbf{z}_{\text{p}} - |\mathbf{b}_{\text{p}}|^2 \approx 2 \text{Re} \{ \mathbf{b}_{\text{p}}^* \odot \mathbf{G}_{\text{p}} \mathbf{h} \}. \quad (106)$$

Define  $\bar{\mathbf{h}} = [\text{Re}\{\mathbf{h}\}^T \quad \text{Im}\{\mathbf{h}\}^T]^T$ , and

$$\mathbf{A}_{\text{p}} = 2 [\text{Re} \{ \text{diag}(\mathbf{b}_{\text{p}}^*) \mathbf{G}_{\text{p}} \}, -\text{Im} \{ \text{diag}(\mathbf{b}_{\text{p}}^*) \mathbf{G}_{\text{p}} \}]. \quad (107)$$

An initial unstructured channel estimate is

$$\hat{\mathbf{h}}^{(0)} = \mathbf{A}_{\text{p}}^{\dagger} (\mathbf{z}_{\text{p}} - |\mathbf{b}_{\text{p}}|^2). \quad (108)$$

---

## Algorithm 2 Joint DOA and Channel-Gain Estimation

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**Require:** Training observations  $\mathbf{z}_{\text{p}}$ , pilots  $\mathbf{x}_{\text{p}}[t]$ , reference vector  $\mathbf{b}_{\text{p}}$ , RIS geometry, feasible parameter region  $\Omega$ , maximum iteration number  $I_{\text{c}}$ , initial damping factor  $\lambda^{(0)}$ , and threshold  $\epsilon_{\text{c}}$ .

**Ensure:** Estimated parameter vector  $\hat{\boldsymbol{\eta}}$  and channel  $\hat{\mathbf{H}}$ .

- 1: Obtain  $\mathbf{H}^{(0)}$  using (108).
  - 2: Obtain coarse DOAs and gains using (109) and (110).
  - 3: Form  $\boldsymbol{\eta}^{(0)}$  according to (78).
  - 4: **for**  $i = 0, 1, \dots, I_{\text{c}} - 1$  **do**
  - 5:   Compute  $\mathbf{u}_{\text{p}}^{(i)} = \mathbf{g}_{\text{p}}(\boldsymbol{\eta}^{(i)}) + \mathbf{b}_{\text{p}}$ .
  - 6:   Compute the residual  $\mathbf{e}_{\text{p}}^{(i)} = \mathbf{z}_{\text{p}} - \left| \mathbf{u}_{\text{p}}^{(i)} \right|^2$ .
  - 7:   Construct  $\mathbf{J}_{\text{p}}^{(i)}$  using (95).
  - 8:   Compute  $\Delta \boldsymbol{\eta}^{(i)}$  using (100).
  - 9:   Compute the projected trial point using (101).
  - 10:   **if** the trial point decreases  $F_{\text{p}}$  **then**
  - 11:     Accept  $\boldsymbol{\eta}^{(i+1)} = \boldsymbol{\eta}_{\text{trial}}^{(i+1)}$ .
  - 12:     Reduce the damping parameter.
  - 13:   **else**
  - 14:     Set  $\boldsymbol{\eta}^{(i+1)} = \boldsymbol{\eta}^{(i)}$ .
  - 15:     Increase the damping parameter.
  - 16:   **end if**
  - 17:   **if**  $\|\Delta \boldsymbol{\eta}^{(i)}\|_2 / (\|\boldsymbol{\eta}^{(i)}\|_2 + \epsilon_0) < \epsilon_{\text{c}}$  **then**
  - 18:     **break**
  - 19:   **end if**
  - 20: **end for**
  - 21: Set  $\hat{\boldsymbol{\eta}} = \boldsymbol{\eta}^{(i+1)}$ .
  - 22: Reconstruct  $\hat{\mathbf{H}} = \mathbf{H}(\hat{\boldsymbol{\eta}})$ .
- 

After reconstructing  $\mathbf{H}^{(0)}$ , the initial DOA of user  $l$  can be obtained through a coarse angular search:

$$\left( \theta_l^{(0)}, \varphi_l^{(0)} \right) = \arg \max_{(\theta, \varphi) \in \mathcal{G}_{\Omega}} \frac{\left| \mathbf{v}^H(\theta, \varphi) \mathbf{h}_l^{(0)} \right|^2}{\left\| \mathbf{v}(\theta, \varphi) \right\|_2^2}, \quad (109)$$

where  $\mathcal{G}_{\Omega}$  is a coarse angular grid and  $\mathbf{h}_l^{(0)}$  is the  $l$ -th column of  $\mathbf{H}^{(0)}$ . The corresponding path-gain initialization is

$$\alpha_l^{(0)} = \frac{\mathbf{v}^H \left( \theta_l^{(0)}, \varphi_l^{(0)} \right) \mathbf{h}_l^{(0)}}{\left\| \mathbf{v} \left( \theta_l^{(0)}, \varphi_l^{(0)} \right) \right\|_2^2}. \quad (110)$$

If the reference-dominated approximation is inaccurate, multiple coarse-grid candidates or random restarts can be used, and the candidate producing the smallest training residual can be selected. This residual-based selection is important because the intensity objective may contain local minima when the initial point is outside the locally stable region characterized by  $\mathbf{J}_{\text{p}}$ .

The complete joint estimation procedure is summarized in Algorithm 2.

The Gauss–Newton method is a local optimization procedure and does not guarantee convergence to the global minimizer from an arbitrary initialization. Its use is justified here by the differentiable low-dimensional channel parameterization, the availability of pilot observations, and the possibility

of obtaining a coarse angular initialization before iterative refinement.

### C. Finite-Alphabet Symbol Detection

After the training stage, the estimated channel is used to construct  $\hat{\mathbf{G}}_d$ . The symbol detection problem is based on

$$\mathbf{z}_d = \left| \hat{\mathbf{G}}_d \mathbf{s}_d + \mathbf{b}_d \right|^2 + \mathbf{w}_d, \quad \mathbf{s}_d \in \mathcal{S}^{D_s}. \quad (111)$$

Unlike the training-stage unknowns, the transmitted symbols belong to a finite constellation. Therefore, the relevant theoretical quantity is the energy-domain separation between candidate symbol vectors rather than the full-rank condition of a continuous Jacobian. A detector can be reliable only when the estimated mapping assigns distinct and sufficiently separated intensity vectors to different constellation points. This motivates using ML detection as a benchmark and using the same distance metric to initialize and evaluate low-complexity iterative detectors.

1) *Maximum-Likelihood Detection*: Assuming independent Gaussian post-detection noise with identical variance, the maximum-likelihood detector is equivalent to

$$\hat{\mathbf{s}}_{\text{ML}} = \arg \min_{\mathbf{s} \in \mathcal{S}^{D_s}} \left\| \mathbf{z}_d - \left| \hat{\mathbf{G}}_d \mathbf{s} + \mathbf{b}_d \right|^2 \right\|_2^2. \quad (112)$$

For two candidates  $\mathbf{s}_i$  and  $\mathbf{s}_j$ , define

$$\hat{d}_E(\mathbf{s}_i, \mathbf{s}_j) = \left| \left| \hat{\mathbf{G}}_d \mathbf{s}_i + \mathbf{b}_d \right|^2 - \left| \hat{\mathbf{G}}_d \mathbf{s}_j + \mathbf{b}_d \right|^2 \right|_2. \quad (113)$$

The corresponding minimum energy-domain distance is

$$\hat{d}_{\min, d}^{(E)} = \min_{\mathbf{s}_i \neq \mathbf{s}_j} \hat{d}_E(\mathbf{s}_i, \mathbf{s}_j). \quad (114)$$

The ML detector provides a direct connection between the identifiability theory in Section III and the simulated symbol error rate. In particular, a positive  $\hat{d}_{\min, d}^{(E)}$  ensures that different candidate symbol vectors produce different noiseless intensity observations under the estimated model.

However, exhaustive ML detection requires evaluating  $|\mathcal{S}|^{D_s}$  candidate vectors. Its complexity grows exponentially with the detected block length, making it primarily suitable for short blocks and performance benchmarking.

2) *Linearized Initialization and Candidate Screening*: The reference term also provides a simple way to initialize the data detector. If the reference-induced interference term dominates the unknown self-intensity over the current block, then

$$\mathbf{z}_d - |\mathbf{b}_d|^2 \approx 2 \operatorname{Re}\{\mathbf{b}_d^* \odot \hat{\mathbf{G}}_d \mathbf{s}_d\}. \quad (115)$$

With

$$\bar{\mathbf{s}} = [\operatorname{Re}\{\mathbf{s}\}^T \quad \operatorname{Im}\{\mathbf{s}\}^T]^T, \quad (116)$$

define the real-valued linearized data matrix

$$\mathbf{A}_d = 2 [\operatorname{Re}\{\operatorname{diag}(\mathbf{b}_d^*) \hat{\mathbf{G}}_d\} \quad -\operatorname{Im}\{\operatorname{diag}(\mathbf{b}_d^*) \hat{\mathbf{G}}_d\}]. \quad (117)$$

An initial continuous estimate is

$$\bar{\mathbf{s}}_{\text{lin}} = (\mathbf{A}_d^T \mathbf{A}_d + \gamma_{\text{lin}} \mathbf{I})^{-1} \mathbf{A}_d^T (\mathbf{z}_d - |\mathbf{b}_d|^2), \quad (118)$$

where  $\gamma_{\text{lin}} \geq 0$  is a small regularization parameter. The corresponding quantized vector is obtained after converting  $\bar{\mathbf{s}}_{\text{lin}}$  back to its complex form, denoted by  $\mathbf{s}_{\text{lin}, \text{cont}}$ :

$$\mathbf{s}_{\text{lin}} = \mathcal{Q}_{\mathcal{S}}(\mathbf{s}_{\text{lin}, \text{cont}}) \quad (119)$$

This vector is used either as the initial point of an iterative detector or as the center of a reduced candidate list. For example, one may enumerate only the constellation vectors whose Hamming distance from  $\mathbf{s}_{\text{lin}}$  is no larger than a small integer. This screened ML detector is no longer globally optimal, but it often captures most of the ML gain when  $\hat{d}_{\min, d}^{(E)}$  is not too small.

3) *Reference-Assisted Gerchberg–Saxton Detection*: The conventional Gerchberg–Saxton algorithm alternates between a measurement-magnitude constraint and a signal-domain constraint. The detector considered here is not a direct application of the conventional homogeneous phase-retrieval algorithm. Instead, it is adapted to the affine observation model by including the known reference vector in the forward projection and subtracting it before the signal-domain back-projection.

Define the nonnegative measured-magnitude vector as

$$\mathbf{r}_d = \sqrt{[\mathbf{z}_d]_+}, \quad (120)$$

where

$$[\mathbf{z}]_+ = \max(\mathbf{z}, \mathbf{0}) \quad (121)$$

is applied element-wise to avoid taking the square root of negative noisy measurements.

Given the current symbol estimate  $\mathbf{s}^{(i)}$ , the predicted affine field is

$$\mathbf{u}_d^{(i)} = \hat{\mathbf{G}}_d \mathbf{s}^{(i)} + \mathbf{b}_d. \quad (122)$$

The projection onto the measured-magnitude set is

$$\tilde{\mathbf{u}}_d^{(i)} = \mathbf{r}_d \odot \exp(j \angle \mathbf{u}_d^{(i)}). \quad (123)$$

The reference vector is then removed, and the resulting field is projected back onto the signal domain. To improve numerical stability when  $\hat{\mathbf{G}}_d$  is ill-conditioned, define the regularized back-projection matrix

$$\mathbf{B}_d = (\hat{\mathbf{G}}_d^H \hat{\mathbf{G}}_d + \gamma \mathbf{I})^{-1} \hat{\mathbf{G}}_d^H, \quad (124)$$

where  $\gamma \geq 0$  is a regularization parameter. When  $\gamma = 0$  and  $\hat{\mathbf{G}}_d$  has full column rank,  $\mathbf{B}_d = \hat{\mathbf{G}}_d^\dagger$ .

The continuous signal-domain update is

$$\tilde{\mathbf{s}}^{(i+1)} = \mathbf{B}_d (\tilde{\mathbf{u}}_d^{(i)} - \mathbf{b}_d). \quad (125)$$

The finite-alphabet structure is incorporated through

$$\mathbf{s}^{(i+1)} = \mathcal{Q}_{\mathcal{S}}(\tilde{\mathbf{s}}^{(i+1)}), \quad (126)$$

where  $\mathcal{Q}_{\mathcal{S}}(\cdot)$  denotes element-wise nearest-neighbor quantization onto the constellation  $\mathcal{S}$ .

The projection in (126) may be applied after every iteration or only after the continuous iterations converge. Applying it at every iteration exploits the finite-alphabet prior more strongly but may cause premature stagnation when the current estimate is far from the transmitted symbol. Applying it only at the final iteration generally produces a smoother continuous iteration trajectory.

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**Algorithm 3** Reference-Assisted Gerchberg–Saxton Symbol Detection
 

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**Require:** Data observations  $\mathbf{z}_d$ , matrix  $\hat{\mathbf{G}}_d$ , reference vector  $\mathbf{b}_d$ , constellation  $\mathcal{S}$ , regularization factor  $\gamma$ , maximum iteration number  $I_d$ , and threshold  $\epsilon_d$ .

**Ensure:** Detected symbol vector  $\hat{\mathbf{s}}_d$ .

- 1: Compute  $\mathbf{r}_d = \sqrt{[\mathbf{z}_d]_+}$ .
  - 2: Precompute  $\mathbf{B}_d$  according to (124).
  - 3: Initialize  $\mathbf{s}^{(0)}$  using a linearized estimate, the previous data block, or a random constellation vector.
  - 4: **for**  $i = 0, 1, \dots, I_d - 1$  **do**
  - 5:   Compute the predicted affine field using (122).
  - 6:   Enforce the measured-magnitude constraint using (123).
  - 7:   Back-project using (125).
  - 8:   Apply the constellation projection using (126).
  - 9:   **if**  $\|\mathbf{s}^{(i+1)} - \mathbf{s}^{(i)}\|_2 / (\|\mathbf{s}^{(i)}\|_2 + \epsilon_0) < \epsilon_d$  **then**
  - 10:     **break**
  - 11:   **end if**
  - 12: **end for**
  - 13: Output  $\hat{\mathbf{s}}_d = \mathcal{Q}_{\mathcal{S}}(\mathbf{s}^{(i+1)})$ .
- 

The complete detector is summarized in Algorithm 3.

The proposed reference-assisted GS detector is attractive for repeated data detection because  $\mathbf{B}_d$  can be precomputed whenever the estimated channel remains unchanged over a coherence block. Each subsequent iteration then mainly involves matrix-vector multiplications and element-wise operations.

Nevertheless, the GS detector remains a nonconvex alternating-projection algorithm. Its convergence can depend on the initialization, the conditioning of  $\hat{\mathbf{G}}_d$ , the reference-wave design, and the noise level. Therefore, it is evaluated against the ML benchmark rather than claimed to be globally optimal. In simulations, multiple initializations can be used, including  $\mathbf{s}_{\text{lin}}$ , the previous detected block, and random constellation vectors. The final output is selected as the candidate with the smallest intensity residual

$$\left\| \mathbf{z}_d - \left| \hat{\mathbf{G}}_d \mathbf{s} + \mathbf{b}_d \right|^2 \right\|_2^2. \quad (127)$$

This selection rule is consistent with ML detection, but it avoids exhaustive enumeration over the whole constellation block.

4) *Optional Projected Gauss–Newton Detector:* For comparison, the finite-alphabet constraint can also be relaxed temporarily, resulting in

$$\hat{\mathbf{s}}_{\text{cont}} = \arg \min_{\mathbf{s} \in \mathbb{C}^{D_s}} \frac{1}{2} \left\| \mathbf{z}_d - \left| \hat{\mathbf{G}}_d \mathbf{s} + \mathbf{b}_d \right|^2 \right\|_2^2. \quad (128)$$

Define  $\bar{\mathbf{s}} = [\text{Re}\{\mathbf{s}\}^T \quad \text{Im}\{\mathbf{s}\}^T]^T$ . At iteration  $i$ , let

$$\mathbf{u}_d^{(i)} = \hat{\mathbf{G}}_d \mathbf{s}^{(i)} + \mathbf{b}_d, \quad (129)$$

and

$$\mathbf{e}_d^{(i)} = \mathbf{z}_d - \left| \mathbf{u}_d^{(i)} \right|^2. \quad (130)$$

The real-valued Jacobian is

$$\mathbf{J}_d^{(i)} = 2 \left[ \text{Re}\{\text{diag}((\mathbf{u}_d^{(i)})^*) \hat{\mathbf{G}}_d\}, -\text{Im}\{\text{diag}((\mathbf{u}_d^{(i)})^*) \hat{\mathbf{G}}_d\} \right]. \quad (131)$$

The damped update is

$$\Delta \bar{\mathbf{s}}^{(i)} = [(\mathbf{J}_d^{(i)})^T \mathbf{J}_d^{(i)} + \mu^{(i)} \mathbf{I}]^{-1} (\mathbf{J}_d^{(i)})^T \mathbf{e}_d^{(i)}. \quad (132)$$

The real vector is updated as

$$\bar{\mathbf{s}}^{(i+1)} = \bar{\mathbf{s}}^{(i)} + \Delta \bar{\mathbf{s}}^{(i)}, \quad (133)$$

and then converted back to the complex symbol domain. A projection onto a box containing the constellation may be used during the continuous iterations to avoid unreasonable excursions when the linearized model is poor. After convergence, the continuous estimate is projected onto the constellation:

$$\hat{\mathbf{s}}_{\text{PGN}} = \mathcal{Q}_{\mathcal{S}}(\mathbf{s}_{\text{cont}}). \quad (134)$$

The projected Gauss–Newton detector is included as an optional algorithmic baseline. Compared with the reference-assisted GS detector, it generally requires more expensive Jacobian construction and linear-system solution at every iteration, but it may converge in fewer iterations when initialized sufficiently close to a desirable solution.

#### D. Computational Complexity and Implementation Considerations

Let  $I_c$  and  $I_d$  denote the numbers of channel-estimation and symbol-detection iterations, respectively. The principal computational costs are summarized in Table I.

The cost in Table I separates operations that are repeated every iteration from operations that can be amortized over a coherence block. For joint DOA and gain estimation, the dominant operations are the formation of  $(\mathbf{J}_p)^T \mathbf{J}_p$  and the solution of the  $D_\eta \times D_\eta$  linear system. Because  $D_\eta = 4L$  for the two-dimensional far-field model, the parameter dimension does not increase directly with the number of RIS elements. This represents a substantial reduction compared with unstructured full-channel estimation, whose real-valued dimension is  $2N_r L$ .

For the GS detector, the regularized back-projection matrix in (124) requires a one-time cost of approximately

$$\mathcal{O}(Q_d D_s^2 + D_s^3). \quad (135)$$

This cost is amortized over all data blocks transmitted within the same channel-coherence interval. Once  $\mathbf{B}_d$  has been precomputed, each iteration requires minimal computational effort, primarily involving two matrix-vector multiplications and element-wise magnitude or phase operations.

The linearized detector in (118) has a similar one-time matrix-inversion cost when  $\hat{\mathbf{G}}_d$  and  $\mathbf{b}_d$  are fixed. Therefore, it is useful not only as a standalone low-complexity detector but also as a deterministic initial point for GS or projected GN iterations.

The data detector may also utilize warm-start initialization. If the channel and transmitted waveform vary slowly between adjacent blocks, the estimate from the previous block can

TABLE I: Computational Complexity of the Main Recovery Algorithms

| Method                | Unknown dimension | Approximate complexity                       | Primary role                           |
|-----------------------|-------------------|----------------------------------------------|----------------------------------------|
| Joint damped GN       | $D_\eta$          | $\mathcal{O}(I_c (Q_p D_\eta^2 + D_\eta^3))$ | DOA and channel-gain estimation        |
| Exhaustive ML         | $D_s$             | $\mathcal{O}( S ^{D_s} Q_d D_s)$             | Performance benchmark                  |
| Linearized detector   | $D_s$             | $\mathcal{O}(Q_d D_s^2 + D_s^3)$             | Initialization and candidate screening |
| Reference-assisted GS | $D_s$             | $\mathcal{O}(I_d Q_d D_s)$                   | Low-complexity symbol detection        |
| Projected GN          | $2D_s$            | $\mathcal{O}(I_d (Q_d D_s^2 + D_s^3))$       | Optional detection baseline            |

serve as  $\mathbf{s}^{(0)}$ . For independently modulated blocks, a reference-dominated linearized estimate or multiple random constellation initializations may be employed.

The stopping conditions should include a small positive constant  $\epsilon_0$  in the denominator to avoid numerical instability when the current estimate has a small norm. In addition to the relative-update criterion used in Algorithms 2 and 3, a residual-based condition can be employed:

$$\frac{\|\mathbf{z} - |\mathbf{G}\boldsymbol{\xi}^{(i)} + \mathbf{b}|^2\|_2}{\|\mathbf{z}\|_2 + \epsilon_0} < \epsilon_r, \quad (136)$$

where  $\boldsymbol{\xi}^{(i)}$  denotes the current channel-parameter or symbol estimate.

#### E. Impact of Channel Estimation Errors

The data-phase detector constructs its candidate intensity patterns using the estimated channel. Therefore, channel estimation errors introduce a model mismatch in addition to the post-detection noise.

Let

$$\hat{\mathbf{H}} = \mathbf{H} + \Delta\mathbf{H}, \quad (137)$$

and

$$\hat{\mathbf{G}}_d = \mathbf{G}_d + \Delta\mathbf{G}_d. \quad (138)$$

Define the true and mismatched intensity mappings as

$$\mathcal{T}_d(\mathbf{s}) = |\mathbf{G}_d \mathbf{s} + \mathbf{b}_d|^2, \quad (139)$$

and

$$\hat{\mathcal{T}}_d(\mathbf{s}) = |\hat{\mathbf{G}}_d \mathbf{s} + \mathbf{b}_d|^2. \quad (140)$$

The intensity-domain mismatch for a candidate  $\mathbf{s}$  is

$$\Delta\mathbf{z}_{\text{CSI}}(\mathbf{s}) = \hat{\mathcal{T}}_d(\mathbf{s}) - \mathcal{T}_d(\mathbf{s}). \quad (141)$$

Let

$$\mathbf{u}_d(\mathbf{s}) = \mathbf{G}_d \mathbf{s} + \mathbf{b}_d, \quad (142)$$

and

$$\Delta\mathbf{u}_d(\mathbf{s}) = \Delta\mathbf{G}_d \mathbf{s}. \quad (143)$$

The exact mismatch can be expanded as

$$\Delta\mathbf{z}_{\text{CSI}}(\mathbf{s}) = 2 \operatorname{Re}\{\mathbf{u}_d^*(\mathbf{s}) \odot \Delta\mathbf{u}_d(\mathbf{s})\} + |\Delta\mathbf{u}_d(\mathbf{s})|^2. \quad (144)$$

For sufficiently small channel errors, the second-order term can be neglected:

$$\Delta\mathbf{z}_{\text{CSI}}(\mathbf{s}) \approx 2 \operatorname{Re}\{(\mathbf{G}_d \mathbf{s} + \mathbf{b}_d)^* \odot \Delta\mathbf{G}_d \mathbf{s}\}. \quad (145)$$

The exact expansion also gives a deterministic upper bound. If

$$\|\Delta\mathbf{G}_d\|_2 \leq \epsilon_G, \quad \|\mathbf{s}\|_2 \leq R_s, \quad (146)$$

then

$$\|\Delta\mathbf{z}_{\text{CSI}}(\mathbf{s})\|_2 \leq 2\|\mathbf{u}_d(\mathbf{s})\|_\infty \epsilon_G R_s + \epsilon_G^2 R_s^2. \quad (147)$$

Furthermore, because  $\Delta\mathbf{G}_d = (\mathbf{I} \otimes \Delta\mathbf{H})\mathbf{\Pi P}_{\text{tx},d}$  for the data-phase composition in (68),

$$\epsilon_G \leq \|\Delta\mathbf{H}\|_2 \|\mathbf{\Pi P}_{\text{tx},d}\|_2. \quad (148)$$

Thus, the end-to-end detector is sensitive not only to the NMSE of the channel estimate but also to the gain of the data-phase waveform mapping.

Suppose that  $\mathbf{s}_d$  is transmitted. The actual observation can be rewritten relative to the mismatched model as

$$\mathbf{z}_d = \hat{\mathcal{T}}_d(\mathbf{s}_d) + \mathbf{e}_{\text{eff}}, \quad (149)$$

where

$$\mathbf{e}_{\text{eff}} = \mathbf{w}_d + \Delta\mathbf{z}_{\text{CSI}}(\mathbf{s}_d). \quad (150)$$

For the nearest-intensity or ML detector, a sufficient condition for correctly distinguishing the transmitted vector from every competing candidate is

$$\|\mathbf{e}_{\text{eff}}\|_2 < \frac{1}{2} \min_{\mathbf{s}' \neq \mathbf{s}_d} \|\hat{\mathcal{T}}_d(\mathbf{s}_d) - \hat{\mathcal{T}}_d(\mathbf{s}')\|_2. \quad (151)$$

Therefore, the energy-domain minimum distance determines the tolerance not only to post-detection noise but also to channel-induced model mismatch. A reference pattern producing a larger minimum distance provides a larger robustness margin under imperfect CSI.

Equivalently, if

$$\epsilon_{\text{CSI}} \triangleq \max_{\mathbf{s} \in \mathcal{S}^{D_s}} \|\Delta\mathbf{z}_{\text{CSI}}(\mathbf{s})\|_2, \quad (152)$$

then a convenient conservative design target is

$$\|\mathbf{w}_d\|_2 + \epsilon_{\text{CSI}} < \frac{\hat{d}_{\min,d}^{(E)}}{2}. \quad (153)$$

This inequality gives a direct engineering interpretation of the third-section stability result: increasing  $\hat{d}_{\min,d}^{(E)}$  increases the admissible sum of receiver noise and CSI-induced intensity mismatch.

For iterative GS or projected GN detection, (151) does not by itself guarantee convergence to the transmitted symbol because these algorithms may converge to local fixed points. Nevertheless, a larger pairwise energy-domain separation generally produces a more favorable detection landscape and reduces the probability that different symbol vectors generate nearly indistinguishable intensity patterns.

The channel-estimation quality is evaluated using

$$\text{NMSE}_H = \frac{\mathbb{E}\{\|\hat{\mathbf{H}} - \mathbf{H}\|_F^2\}}{\mathbb{E}\{\|\mathbf{H}\|_F^2\}}. \quad (154)$$

In Section V, the symbol error rates obtained under perfect CSI and estimated CSI will be compared to quantify the end-to-end effect of training-stage estimation errors.

In summary, the proposed two-stage algorithm assigns different recovery methods according to the structure of the unknown variables. The damped Gauss–Newton method performs accurate local refinement of continuous DOA and channel-gain parameters during the relatively infrequent training stage. Maximum-likelihood detection characterizes the finite-alphabet performance limit, while the reference-assisted Gerchberg–Saxton detector provides a lower-complexity implementation for repeated data-symbol recovery. This task-oriented algorithm selection is consistent with the uniqueness and stability criteria established in Section III.

## V. SIMULATION RESULTS

In this section, numerical simulations are conducted to evaluate the proposed reference-assisted holographic RIS receiver. We first examine the channel estimation performance in the training phase, then investigate the symbol detection performance under both perfect and estimated channel state information (CSI). Finally, we study the impact of reference-wave design and RIS measurement diversity on the energy-domain separability and detection reliability.

### A. Simulation Setup

We consider an uplink multi-user communication system assisted by a holographic RIS receiver. Unless otherwise specified, the RIS receiver is modeled as a uniform planar array placed on the  $yz$ -plane with  $N_y \times N_z$  receiving elements. The total number of RIS elements is

$$N_r = N_y N_z. \quad (155)$$

The inter-element spacings are set as

$$d_y = d_z = \frac{\lambda}{2}, \quad (156)$$

where  $\lambda$  is the carrier wavelength. The carrier frequency is denoted by  $f_c$ , and the corresponding wavelength is  $\lambda = c/f_c$ , where  $c$  is the speed of light.

The far-field channel model in Section II is adopted. The channel matrix is generated as

$$\mathbf{H} = \mathbf{V}\mathbf{\Gamma}, \quad (157)$$

where  $\mathbf{V}$  is the RIS array manifold matrix and  $\mathbf{\Gamma} = \text{diag}(\alpha_1, \dots, \alpha_L)$  contains the complex channel gains of the  $L$  users. The channel gains are modeled as

$$\alpha_l = \sqrt{\beta_l} e^{j\omega_l}, \quad (158)$$

where  $\beta_l$  denotes the large-scale channel power gain and  $\omega_l$  is uniformly distributed over  $[0, 2\pi)$ . The users' angles of arrival are selected according to the considered simulation scenario. In particular, to study the effect of spatial correlation, the angular separation between users will be varied in some simulations.

For the transmitter, each user transmits symbols drawn from a normalized constellation  $\mathcal{S}$ , such as QPSK or 16-QAM. The pulse-shaping matrix is constructed according to the

pulse-shaping filter introduced in Section II. Unless otherwise specified, the data block contains  $K_d$  symbols per user, and the total data-symbol dimension is

$$D_s = LK_d. \quad (159)$$

During the training phase, each user transmits known pilot symbols over  $T_p$  pilot observations.

The received intensity measurements are generated according to the reference-assisted square-law model

$$\mathbf{z} = |\mathbf{G}\boldsymbol{\theta} + \mathbf{b}|^2 + \mathbf{w}, \quad (160)$$

where  $\mathbf{w}$  is modeled as additive real-valued Gaussian noise in the intensity domain. The signal-to-noise ratio (SNR) is defined as

$$\text{SNR} = 10 \log_{10} \left( \frac{\mathbb{E} \left\{ \|\mathbf{G}\boldsymbol{\theta} + \mathbf{b}\|_2^2 \right\}}{\mathbb{E} \left\{ \|\mathbf{w}\|_2^2 \right\}} \right). \quad (161)$$

We compare the following reference-wave designs:

- **No reference:**  $\mathbf{b} = \mathbf{0}$ .
- **Constant reference:** all reference entries have the same phase.
- **Random reference:** the reference phases are independently drawn from  $\mathcal{U}[0, 2\pi)$ .
- **Uniform reference:** the reference phases are uniformly distributed over  $[0, 2\pi)$ .
- **Codebook-optimized reference:** the reference pattern is selected from a finite codebook according to the design criteria in Section IV.

For fair comparison, the total reference power is kept identical among different reference designs unless otherwise stated.

For channel estimation, the damped Gauss–Newton algorithm in Section V is used. The estimation performance is measured by the normalized mean-square error (NMSE)

$$\text{NMSE}_H = \frac{\mathbb{E} \left\{ \|\hat{\mathbf{H}} - \mathbf{H}\|_F^2 \right\}}{\mathbb{E} \left\{ \|\mathbf{H}\|_F^2 \right\}}. \quad (162)$$

For symbol detection, we compare maximum-likelihood (ML) detection, projected Gauss–Newton detection, and biased Gerchberg–Saxton (GS) detection. The detection performance is evaluated using the symbol error rate (SER), defined as

$$\text{SER} = \frac{\text{number of incorrectly detected symbols}}{\text{total number of transmitted symbols}}. \quad (163)$$

To connect the simulations with the theoretical analysis, we also evaluate the energy-domain minimum distance

$$d_{\min}^{(E)} = \min_{\mathbf{s}_i \neq \mathbf{s}_j} \left\| |\mathbf{G}_d \mathbf{s}_i + \mathbf{b}_d|^2 - |\mathbf{G}_d \mathbf{s}_j + \mathbf{b}_d|^2 \right\|_2, \quad (164)$$

as well as the singular-value and condition-number metrics associated with the effective measurement matrix and the Jacobian. All results are averaged over independent channel realizations, noise realizations, and transmitted data blocks.

The default simulation parameters are summarized in Table II. Unless otherwise stated, these default values are used in the following simulations.

TABLE II: Default Simulation Parameters

| Parameter                              | Default value                                 |
|----------------------------------------|-----------------------------------------------|
| Carrier frequency $f_c$                | 3.5 GHz                                       |
| RIS array size                         | $N_y \times N_z = 8 \times 8$                 |
| Number of RIS elements $N_r$           | 64                                            |
| Element spacing                        | $d_y = d_z = \lambda/2$                       |
| Number of users $L$                    | 2 or 4                                        |
| Modulation constellation $\mathcal{S}$ | QPSK / 16-QAM                                 |
| Pilot length $T_p$                     | scenario-dependent                            |
| Data block length $K_d$                | scenario-dependent                            |
| Reference amplitude                    | fixed unless otherwise stated                 |
| Reference phase designs                | none / constant / random / uniform / codebook |
| Channel estimator                      | damped Gauss–Newton                           |
| Symbol detectors                       | ML / projected GN / biased GS                 |
| Performance metrics                    | NMSE / SER / $d_{\min}^{(E)}$                 |

### B. Channel Estimation Performance

In this subsection, we evaluate the performance of the proposed reference-assisted channel estimation scheme in the training phase. The main objective is to verify the role of the reference wave in improving the local identifiability and stability of channel estimation. Unless otherwise specified, the default simulation parameters in Table II are adopted.

We first compare the channel estimation accuracy under different reference-wave designs. The NMSE performance versus SNR is shown in Fig. ???. The compared schemes include no reference, constant-phase reference, random-phase reference, uniform-phase reference, and codebook-optimized reference. This comparison is used to verify whether reference phase diversity can improve the conditioning of the intensity-based channel estimation problem.

Next, we study the effect of the pilot length on channel estimation. According to the local identifiability analysis, the number of intensity observations should be sufficiently large to make the Jacobian full column rank and well conditioned. Fig. ??? shows the NMSE as a function of the pilot length  $T_p$ . This result is intended to demonstrate how additional pilot observations improve the stability of reference-assisted channel estimation.

We then evaluate the influence of reference power. The reference wave provides phase anchoring through the interference term, but its amplitude also affects the dynamic range and the sensitivity of the square-law measurements. Fig. ??? presents the NMSE performance under different reference amplitudes. This figure is used to determine a reasonable reference-power region for the proposed receiver.

To connect the simulation results with the Jacobian-based analysis, we also evaluate the singular-value behavior of the real-valued Jacobian. Fig. ??? shows the minimum singular value and/or condition number of the Jacobian under different reference-wave designs. This result is used to verify that reference patterns leading to larger  $\sigma_{\min}(\mathbf{J}_p)$  or smaller  $\kappa(\mathbf{J}_p)$  also lead to improved NMSE performance.

Finally, we examine the effect of user angular separation. For the far-field channel model, the RIS array manifold matrix  $\mathbf{V}$  becomes ill-conditioned when the users have close angles of arrival. This reduces the separability of the channel parameters and degrades the channel estimation performance. Fig. ??? shows the NMSE versus the angular separation between users. The corresponding condition number of the array manifold

matrix can also be plotted to illustrate the relationship between spatial correlation and estimation accuracy.

Overall, these results are expected to validate the theoretical insights in Sections III and IV: the training reference wave improves channel estimation by enhancing the local sensitivity of the intensity mapping, while the RIS array manifold and pilot structure determine whether the channel parameters are sufficiently distinguishable in the complex field domain.

### C. Symbol Detection Performance with Perfect CSI

In this subsection, we evaluate the data-symbol detection performance under perfect CSI. The purpose is to isolate the impact of reference-assisted intensity measurements and detection algorithms from channel estimation errors. Therefore, the data-phase equivalent measurement matrix  $\mathbf{G}_d$  is constructed using the true channel matrix  $\mathbf{H}$ . Unless otherwise specified, the default simulation parameters in Table II are adopted.

We first compare different symbol detection algorithms under the same reference-wave design. Fig. ??? shows the SER versus SNR for maximum-likelihood (ML) detection, projected Gauss–Newton (GN) detection, and biased Gerchberg–Saxton (GS) detection. The ML detector serves as a benchmark, while projected GN and biased GS provide lower-complexity alternatives. This comparison evaluates the performance-complexity tradeoff of the proposed reference-assisted detection framework.

Next, we investigate the impact of reference-wave design on symbol detection. Fig. ??? compares the SER performance of different reference structures, including no reference, constant-phase reference, random-phase reference, uniform-phase reference, and codebook-optimized reference. For a fair comparison, the total reference power is kept the same for all reference-assisted schemes. This figure is intended to verify that reference phase diversity improves the separability of finite-alphabet symbols in the intensity domain.

To directly validate the energy-domain minimum distance analysis, Fig. ??? presents  $d_{\min}^{(E)}$  under different reference-wave designs. According to the theoretical analysis, a larger  $d_{\min}^{(E)}$  implies stronger pairwise separability after square-law detection. This result is used to explain the SER trends observed in Fig. ???.

We further examine the relationship between the energy-domain minimum distance and the detection performance. Fig. ??? plots the SER as a function of  $d_{\min}^{(E)}$  by varying the reference pattern or reference phase codebook. This figure is used to verify that improving the energy-domain minimum distance leads to a lower symbol error rate, which supports the max-min reference design criterion introduced in Section IV.

Finally, we study the effect of modulation order on the proposed detection framework. Fig. ??? compares the SER performance for different constellations, such as QPSK and 16-QAM. Higher-order modulation leads to denser constellation points and smaller effective pairwise separations, making symbol detection more sensitive to intensity-domain ambiguity and post-detection noise.

Overall, the perfect-CSI results are expected to demonstrate that the proposed reference-assisted receiver can reliably re-

cover finite-alphabet symbols from intensity-only measurements when the reference wave is properly designed. They also show that the energy-domain minimum distance provides a meaningful indicator for predicting and improving symbol detection performance.

#### D. End-to-End Symbol Detection with Estimated CSI

In this subsection, we evaluate the end-to-end data detection performance when the channel is estimated from the training phase. Different from the perfect-CSI results in the previous subsection, the data-phase equivalent measurement matrix is now constructed from the estimated channel  $\hat{\mathbf{H}}$ . Therefore, the detection performance is affected by both post-detection noise and channel estimation error. Unless otherwise specified, the same reference design is used consistently in the training and data phases, and the default simulation parameters in Table II are adopted.

We first compare the SER performance under perfect CSI and estimated CSI. Fig. ?? shows the SER versus SNR for the proposed receiver using different symbol detectors. The perfect-CSI curves provide performance benchmarks, while the estimated-CSI curves quantify the degradation caused by channel estimation error. This comparison validates the necessity of accurate reference-assisted channel estimation for reliable data detection.

Next, we investigate the impact of pilot length on the end-to-end detection performance. Fig. ?? shows the SER as a function of the pilot length  $T_p$ . A larger pilot length generally improves the channel estimate, which in turn reduces the mismatch in the data-phase measurement matrix. This result links the channel estimation accuracy in the training phase to the final symbol detection reliability.

To further quantify the influence of channel estimation error, Fig. ?? plots the SER as a function of the channel estimation NMSE. This can be obtained either by using different pilot lengths and training SNRs or by injecting controlled channel perturbations into the true channel. The purpose of this figure is to verify that a larger channel estimation error leads to a stronger intensity-domain model mismatch and thus degrades symbol detection performance.

We also compare the end-to-end performance under different training reference designs. Fig. ?? shows the SER obtained with no reference, constant-phase reference, random-phase reference, uniform-phase reference, and codebook-optimized reference in the training phase. This comparison demonstrates how the training reference design affects the final data detection performance through the quality of the estimated channel.

Finally, we evaluate the impact of using different reference designs in the data phase under estimated CSI. Fig. ?? compares the SER performance of different data-phase references while using the same channel estimation procedure. This result separates the effect of data-phase symbol separability from the effect of training-phase channel estimation accuracy.

Overall, these results are expected to show that the proposed reference-assisted receiver remains effective under estimated CSI, but its performance depends on both the training-stage channel estimation quality and the data-stage energy-domain

separability. They also verify the analysis in Section V: channel estimation error introduces additional intensity-domain mismatch, while a larger energy-domain minimum distance provides better robustness to this mismatch.

#### E. Effect of Reference and RIS Measurement Diversity

In this subsection, we further investigate how reference diversity and RIS measurement diversity affect the performance of the proposed receiver. The purpose is to verify the design principle developed in Section IV: the effective measurement matrix should preserve the relevant complex-field differences, while the reference wave should project these differences into distinguishable intensity-domain observations. Unless otherwise specified, the default simulation parameters in Table II are adopted.

We first study the impact of the number of RIS receiving elements. Fig. ?? shows the energy-domain minimum distance  $d_{\min}^{(E)}$  as a function of the number of RIS elements  $N_r$ . As the number of RIS elements increases, the receiver obtains more spatial intensity observations, which generally improves the separability of different symbol vectors in the energy domain.

The corresponding SER performance is shown in Fig. ???. This figure is used to verify that the improvement in  $d_{\min}^{(E)}$  obtained from additional RIS observations can be translated into lower symbol detection error rates. Together with Fig. ??, this result highlights the role of the holographic RIS as a spatial measurement device rather than merely a passive aperture.

Next, we evaluate the effect of multiple reference states. Fig. ?? plots  $d_{\min}^{(E)}$  versus the number of reference states  $R$ . Each reference state provides an additional intensity projection of the same underlying complex field. Therefore, increasing  $R$  improves measurement diversity and can reduce the probability that a vulnerable symbol pair remains weakly separated under all reference states.

Fig. ?? further shows the SER performance under different numbers of reference states. This result is intended to demonstrate the tradeoff between detection reliability and measurement overhead. Although more reference states provide stronger intensity-domain diversity, they also require additional measurement time or reference switching operations.

We then examine the effect of user angular separation. For the far-field RIS channel model, smaller angular separation leads to more correlated steering vectors and a more ill-conditioned array manifold matrix  $\mathbf{V}$ . Fig. ?? shows the condition number or minimum singular value of  $\mathbf{V}$  as a function of the user angular separation. This figure directly illustrates how the spatial geometry affects the conditioning of the RIS measurement matrix.

The corresponding end-to-end detection performance is shown in Fig. ???. When the users have close angles of arrival, the steering vectors become highly correlated, which reduces the effective field-domain separation and degrades both channel estimation and symbol detection. As the angular separation increases, the RIS array manifold becomes better conditioned, leading to improved detection reliability.

Finally, we compare reference-only diversity and joint reference-RIS measurement diversity. In the reference-only



case, the effective measurement matrix is fixed while the reference phases are varied. In the joint diversity case, both the reference vector and the RIS measurement configuration are varied across measurement states. Fig. ?? compares the resulting  $d_{\min}^{(E)}$  and/or SER performance. This comparison is used to verify that varying both  $\mathbf{G}$  and  $\mathbf{b}$  provides richer observations than changing the reference vector alone.

Overall, these results are expected to confirm that the proposed receiver benefits from both reference diversity and RIS spatial measurement diversity. The reference wave improves the intensity-domain projection of complex-field differences, while the RIS array and its measurement configurations determine whether these differences are sufficiently preserved before square-law detection. This validates the central design principle of the proposed reference-assisted holographic RIS receiver.

## VI. CONCLUSION

This paper presented a comprehensive framework for signal recovery in holographic RIS receivers using only amplitude measurements. By employing a known reference wave and formulating the recovery as a nonlinear least-squares problem, we enable the estimation of both amplitude and phase of the transmitted signal. The proposed Gauss-Newton algorithm efficiently solves this optimization, with simulations demonstrating accurate recovery under various SNR conditions.

The mathematical derivation provides complete details from signal transmission through the channel, interference with the reference wave at the RIS, intensity measurement, to the final recovery algorithm. This framework serves as a foundation for more advanced RIS receiver designs, including those with adaptive reflection coefficients, multiple users, and frequency-selective channels.

Future work will extend this approach to practical implementations with hardware impairments, investigate deep learning-based recovery for reduced complexity, and explore applications in joint communication and sensing with RIS.

## APPENDIX A

## APPENDIX B

## REFERENCES

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